

SFINCS skillbed: v2.3.0 mt Faber release 2025.02

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1 Introduction

This document reports on the skill of the SFINCS model by Deltares. This document is based on the SFINCS testbed and can be generated (automatically) any time the model code is updated. The SFINCS testbed is an environment, which automatically executes simulations, analyses simulations and generates reports with the results. The SFINCS source code is hosted on a Github repository, which is publicly available (open source). Ideally, every new commit to this repository starts the trigger to create this report. The date and revision number of the SFINCS model code used in this report is given on the cover of this skillbed report.

The function of this skillbed report is twofold. The validation cases show the performance of the SFINCS model for multiple applications in terms of quantitative statistical scores and visualizations. Apart from the performance, this report can also be used to track the code development and verify the impact of a commit in the source code. In this report the results are not discussed since the results can change between different versions of SFINCS.

In the different testcases there is compared between a stable (pre-release) version of SFINCS, the latest release of SFINCS and measurements/observations where available. Small differences between the current and older versions of SFINCS can exist, not directly indicating that the performance of the current model is worse.

The source code of SFINCS is publicly available on GitHub:

<https://github.com/Deltares/SFINCS>

For more information about SFINCS and how to get the executable see:

<https://www.deltares.nl/en/software/sfincs/>

For an elaborate user manual of how to use the SFINCS model see:

<https://sfincs.readthedocs.io/en/latest/>

1.1 Reader's guide

At first, the implementation of basic features (e.g. thin dams and weirs) are tested and discussed in [chapter 2](#). Secondly, in [chapter 3](#), conceptual tests are used to test the performance and numerical implementation of different processes (e.g. flow and waves) in the SFINCS model. Lastly, in order to assess whether the conclusions of the conceptual tests also apply in more realistic cases, case studies are performed and discussed in [chapter 4](#).

2 Implementation

In this chapter, the implementation of simple features (e.g. thin dams and weirs) is tested and discussed.

Test	Short description
Thin dam and weir	Unit test for thin dams and weirs
Thin dam and culvert	Unit test for thin dams and culverts
Constant uniform infiltration	Unit test for 'qinf' infiltration
Constant varying infiltration	Unit test for 'qinffile' infiltration
Curve Number infiltration method A	Unit test for CN without recovery infiltration
Curve Number infiltration method B	Unit test for CN with recovery infiltration
Green	Ampt infiltration method
Unit test for G	A infiltration
Modified Horton infiltration method	Unit test for Horton infiltration
Drainage discharge culvert	Unit test for discharge through drainage culvert
Drainage discharge pump	Unit test for discharge through drainage pump
Storage volume regular	Unit test for storage volume for regular subgrid model
Storage volume quadtree	Unit test for storage volume for quadtree subgrid model
Controlled gates	Unit test for conditionally open gates for sill and zmin

Table 2.1: Overview of the implementation tests in this report.

2.1 Thin dam and weir

Experiment description

In this experiment, the implementation of features like thin dams and weirs into SFINCS are tested.

Results

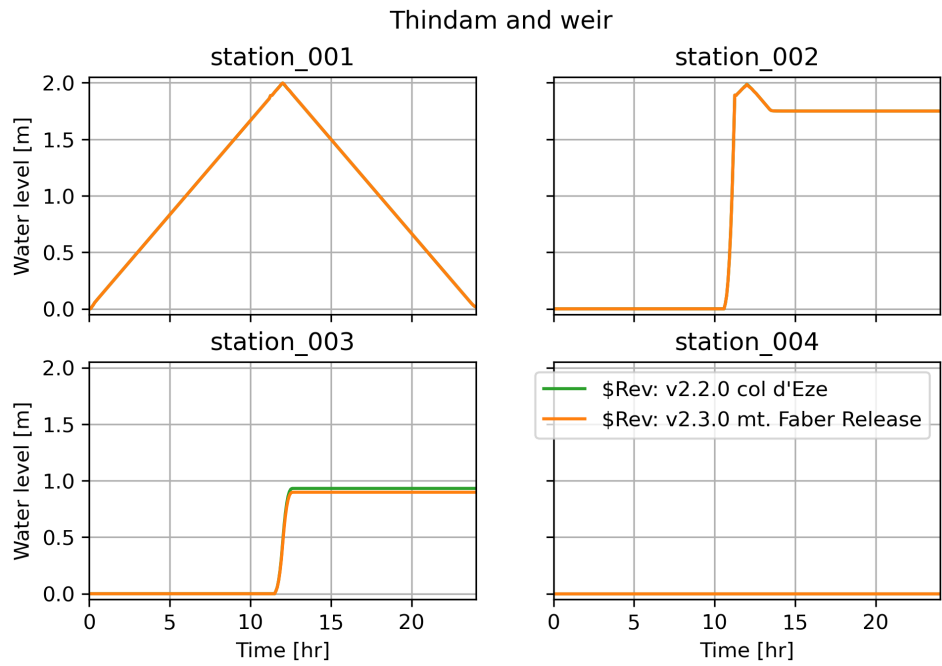


Figure 2.1: Comparison of modelled water levels between latest release and stable release.

Table 2.2: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	3.392	1.460
RMSE loc:station_001	-	0.000
RMSE loc:station_002	-	0.001
RMSE loc:station_003	-	0.024
RMSE loc:station_004	-	0.000

2.2 Thin dam and culvert

Experiment description

In this experiment, the implementation of features like thin dams and a culvert into SFINCS are tested.

Results

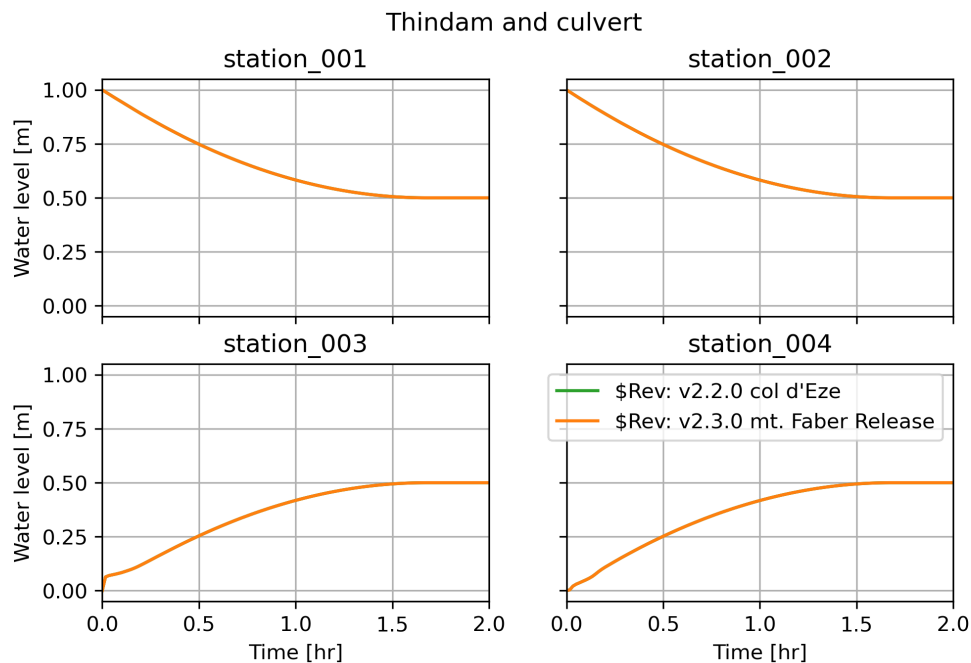


Figure 2.2: Comparison of modelled water levels between latest release and stable release.

Table 2.3: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	0.270	0.168
RMSE loc:station_001	-	0.000
RMSE loc:station_002	-	0.000
RMSE loc:station_003	-	0.000
RMSE loc:station_004	-	0.000

2.3 Constant uniform infiltration

Experiment description

In this experiment, the implementation the spatially uniform constant infiltration of SFINCS is tested.

Results

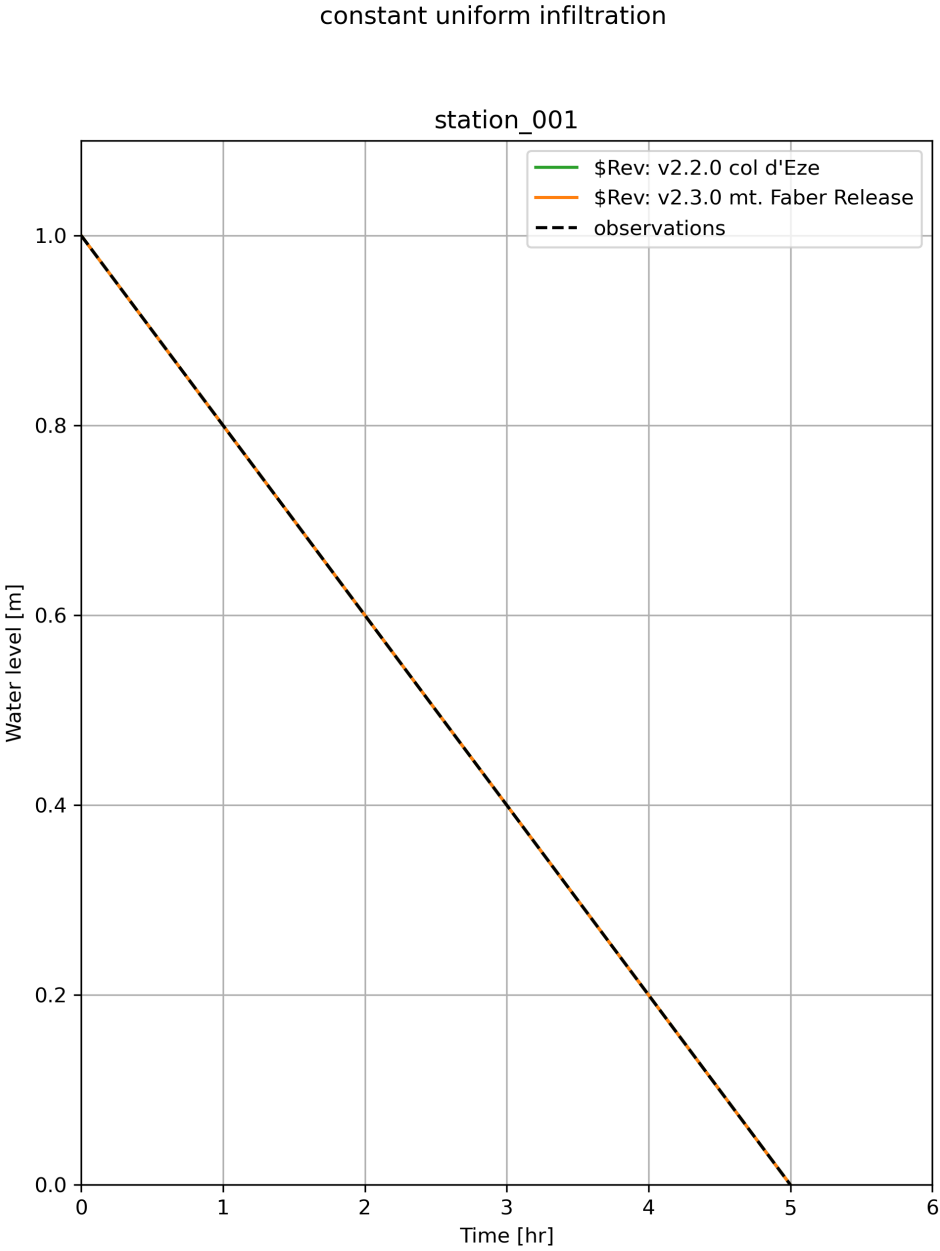


Figure 2.3: Comparison of modelled water levels between latest release and expected result.

Table 2.4: Comparison in runtime and modelled water levels between latest release and expected result.

Release version	Total runtime [s]	RMSE loc:station_001
\$Rev: v2.2.0 col d'Eze	19.622	-
\$Rev: v2.3.0 mt. Faber Release	15.646	0.000

2.4 Constant varying infiltration

Experiment description

In this experiment, the implementation the spatially varying constant infiltration of SFINCS is tested.

Results

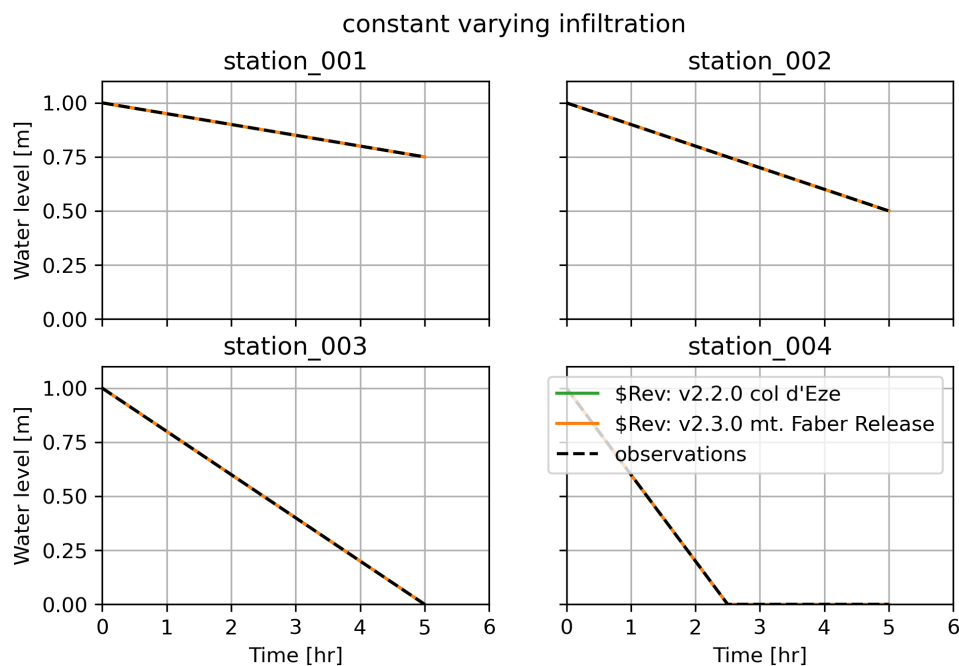


Figure 2.4: Comparison of modelled water levels between latest release and expected result.

Table 2.5: Comparison in runtime and modelled water levels between latest release and expected result.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	42.349	44.055
RMSE loc:station_001	-	0.000
RMSE loc:station_002	-	0.000
RMSE loc:station_003	-	0.000
RMSE loc:station_004	-	0.000

2.5 Curve Number infiltration method - without recovery

Experiment description

In this experiment, the implementation of the Curve Number method A infiltration of SFINCS is tested. The model domain contains a square box with a constant rainfall input. There can be seen that initially, part of the incoming rainfall is infiltrated in about the first 3 hours. Afterwards the soil is saturated, and all rainfall is considered runoff, and the water level increases linearly.

Results

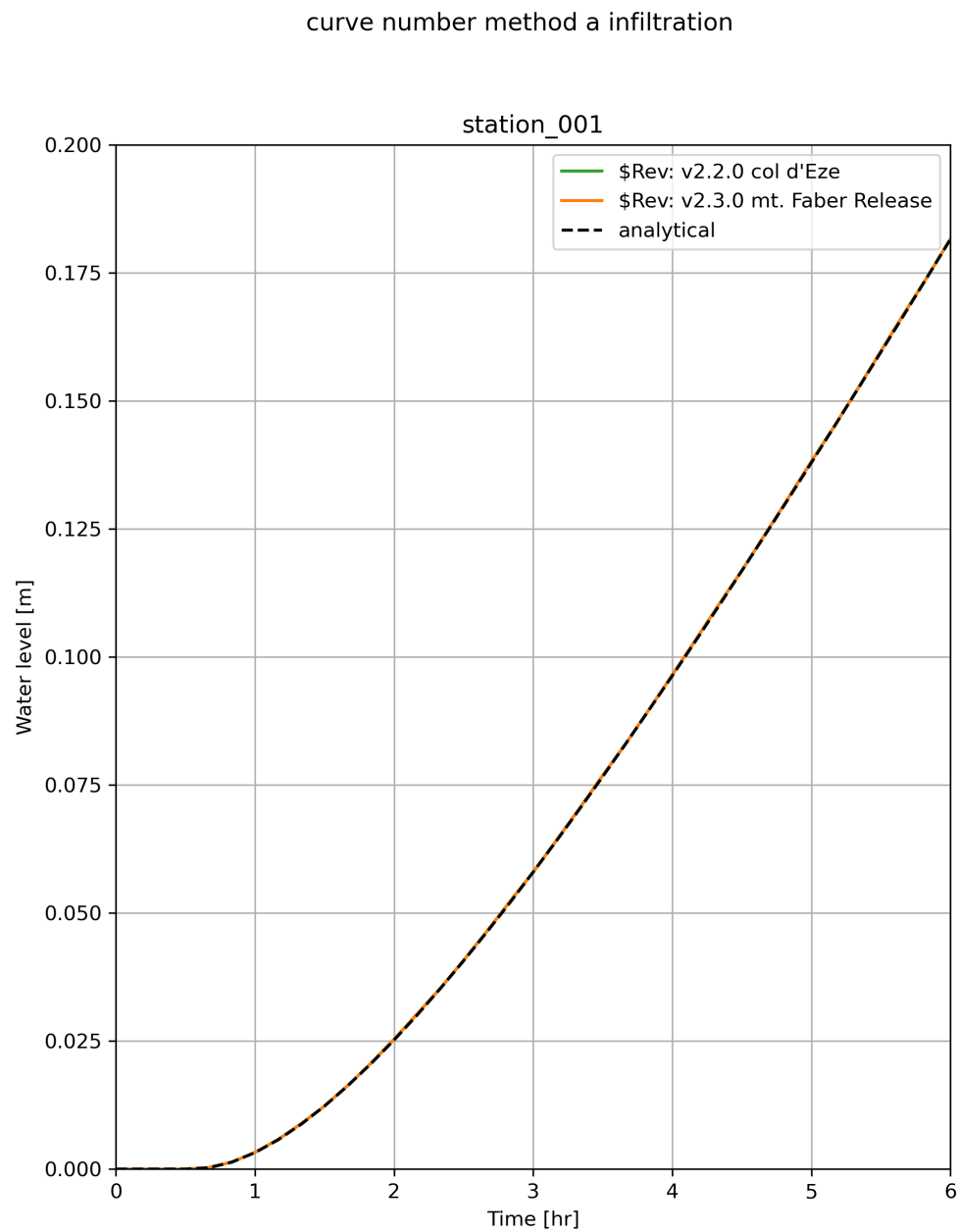


Figure 2.5: Comparison of modelled water levels between latest release and stable version.

Table 2.6: Comparison in runtime and modelled water levels between latest release and stable version.

Release version	Total runtime [s]	RMSE loc:station_001
\$Rev: v2.2.0 col d'Eze	0.086	-
\$Rev: v2.3.0 mt. Faber Release	0.070	0.000

2.6 Curve Number infiltration method - with recovery

Experiment description

In this experiment, the implementation of the Curve Number method B infiltration of SFINCS is tested, being Curve number with recovery. The model domain contains a square box with a varying in time rainfall input. There can be seen that initially, part of the incoming rainfall is infiltrated in about the first 3 hours. Afterwards the soil is saturated, and all rainfall is considered runoff, and the water level increases linearly. Then the rain stops, letting the soil recover and the soil be less saturated, which can be seen during the following rainfall peak.

Results

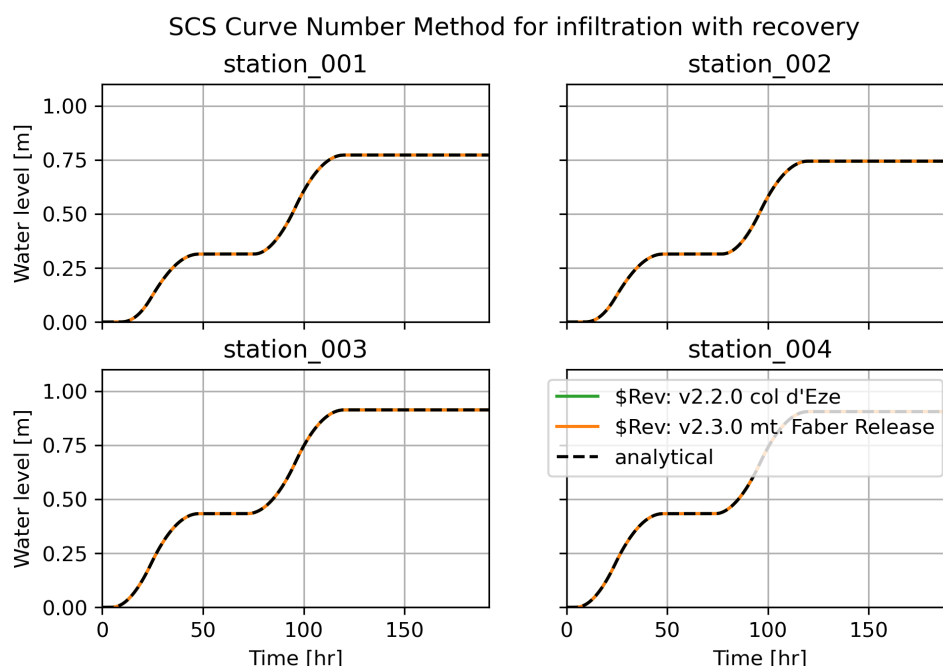


Figure 2.6: Comparison of modelled water levels between latest release and stable version.

Table 2.7: Comparison in runtime and modelled water levels between latest release and stable version.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	120.140	104.003
RMSE loc:station_001	-	0.000
RMSE loc:station_002	-	0.000
RMSE loc:station_003	-	0.000
RMSE loc:station_004	-	0.000

2.7 Green Ampt infiltration method

Experiment description

In this experiment, the implementation of the GreenAmpt infiltration of SFINCS is tested. The model domain contains a square box with a varying in time rainfall input. There can be seen that initially, part of the incoming rainfall is infiltrated in about the first 3 hours. Afterwards the

soil is saturated, and all rainfall is considered runoff, and the water level increases linearly. Then the rain stops, letting the soil recover and the soil be less saturated, which can be seen during the following rainfall peak.

Results

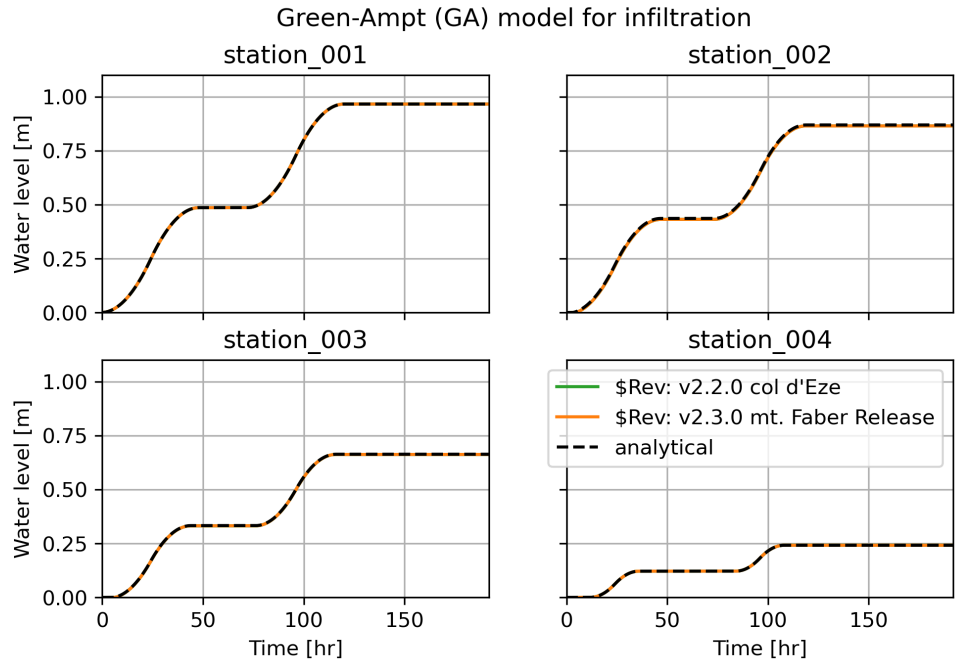


Figure 2.7: Comparison of modelled water levels between latest release and stable version.

Table 2.8: Comparison in runtime and modelled water levels between latest release and stable version.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	56.965	70.140
RMSE loc:station_001	-	0.000
RMSE loc:station_002	-	0.000
RMSE loc:station_003	-	0.000
RMSE loc:station_004	-	0.000

2.8 Modified Horton infiltration method

Experiment description

In this experiment, the implementation of the Modified Horton infiltration of SFINCS is tested. The model domain contains a square box with a varying in time rainfall input. There can be seen that initially, part of the incoming rainfall is infiltrated in about the first 3 hours. Afterwards the soil is saturated, and all rainfall is considered runoff, and the water level increases linearly. Then the rain stops, letting the soil recover and the soil be less saturated, which can be seen during the following rainfall peak.

Results

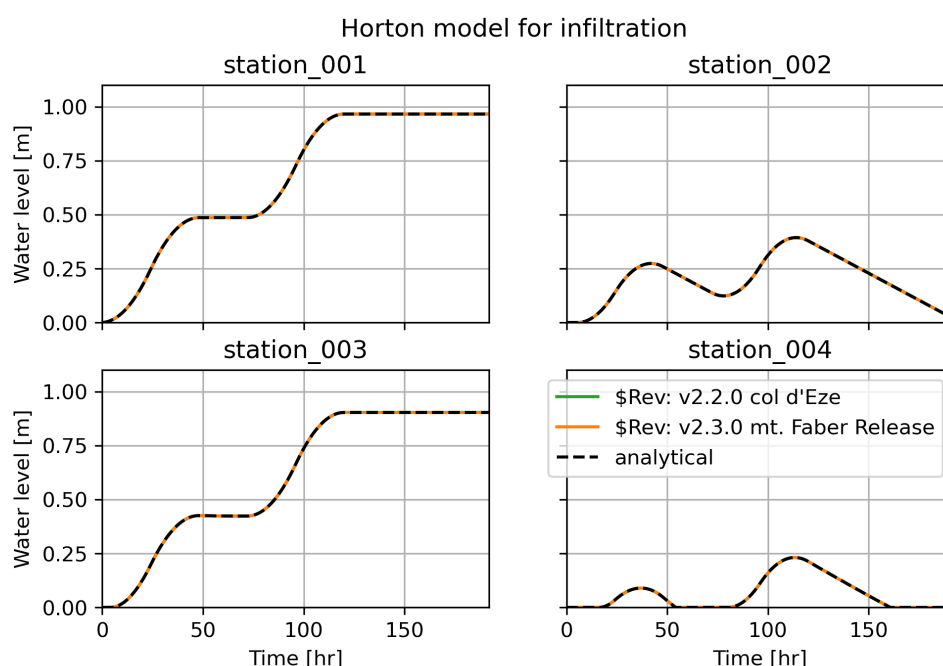


Figure 2.8: Comparison of modelled water levels between latest release and stable version.

Table 2.9: Comparison in runtime and modelled water levels between latest release and stable version.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	53.876	61.841
RMSE loc:station_001	-	0.000
RMSE loc:station_002	-	0.000
RMSE loc:station_003	-	0.000
RMSE loc:station_004	-	0.000

2.9 Drainage culvert

Experiment description

In this experiment, the implementation of features like of keeping track of the discharge through culverts in SFINCS are tested.

Results

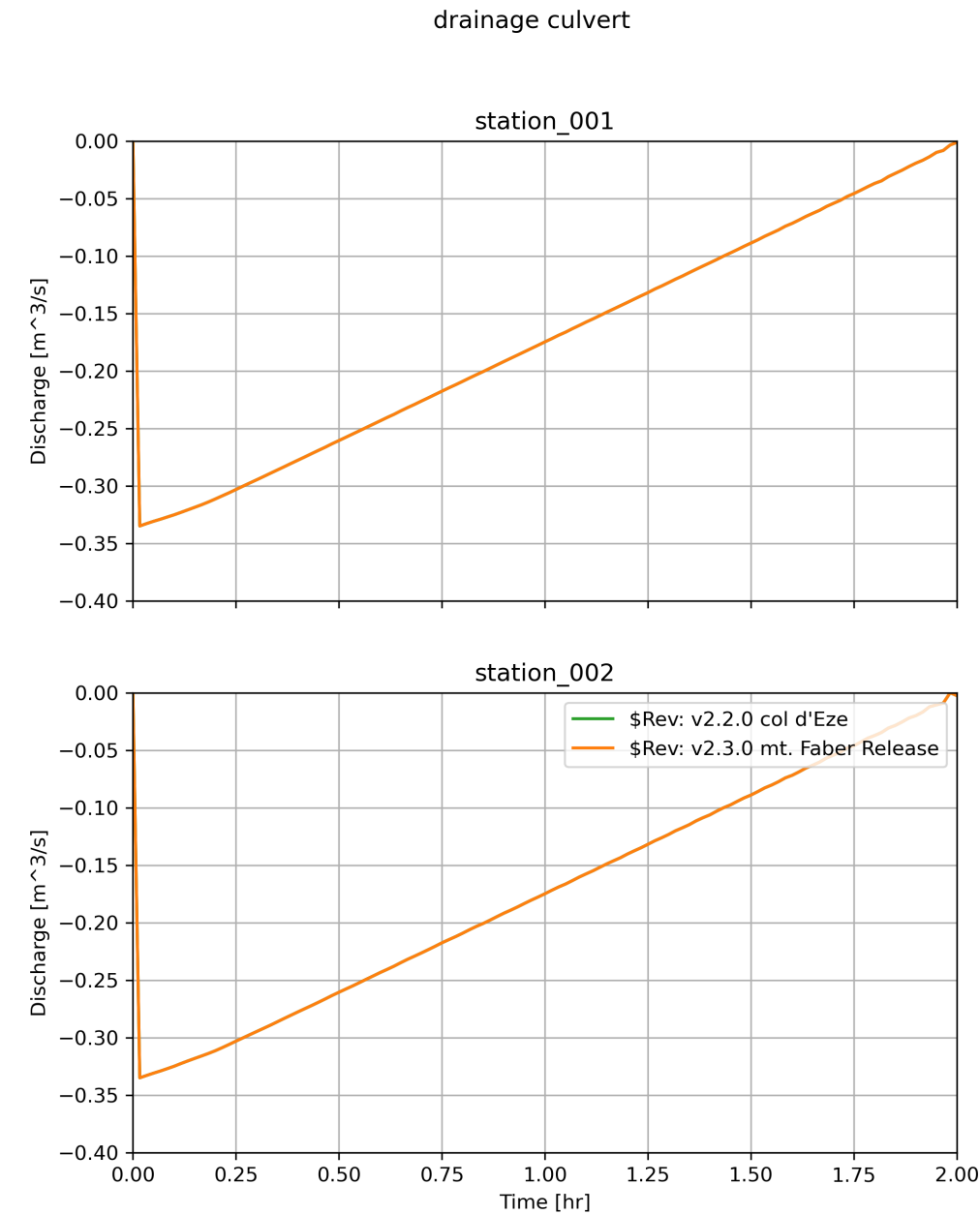


Figure 2.9: Comparison of modelled discharge between latest release and stable release.

Table 2.10: Comparison in runtime and modelled discharge between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:station_001	RMSE loc:station_002
\$Rev: v2.2.0 col d'Eze	0.281	-	-
\$Rev: v2.3.0 mt. Faber Release	0.160	0.000	0.000

2.10 Drainage pump

Experiment description

In this experiment, the implementation of features like of keeping track of the discharge through pump in SFINCS are tested.

Results

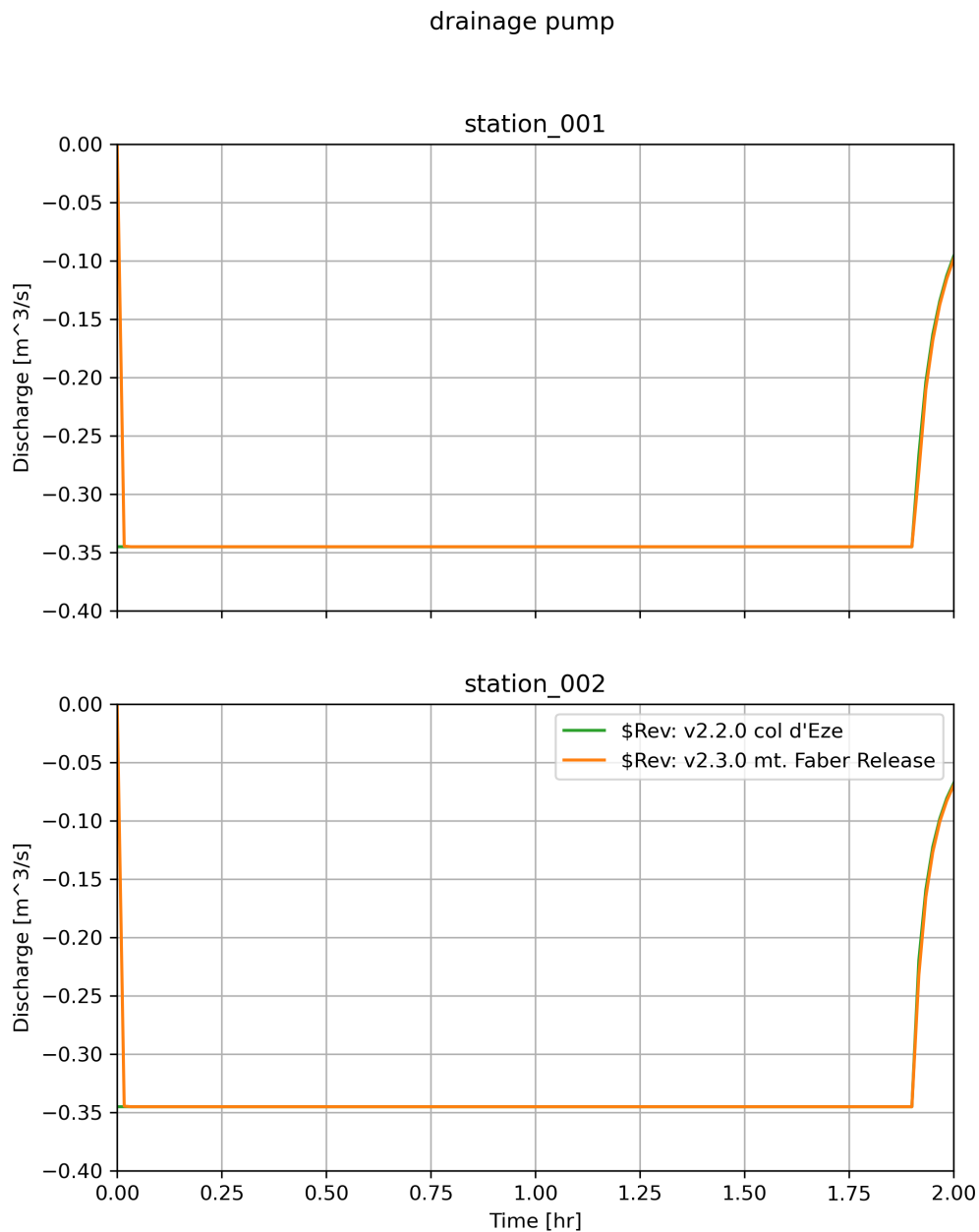


Figure 2.10: Comparison of modelled discharge between latest release and stable release.

Table 2.11: Comparison in runtime and modelled discharge between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:station_001	RMSE loc:station_002
\$Rev: v2.2.0 col d'Eze	0.378	-	-
\$Rev: v2.3.0 mt. Faber Release	0.213	0.031	0.031

2.11 Storage volume

Experiment description

In this experiment, the implementation of features like storage volume for regular subgrid in SFINCS are tested.

Results

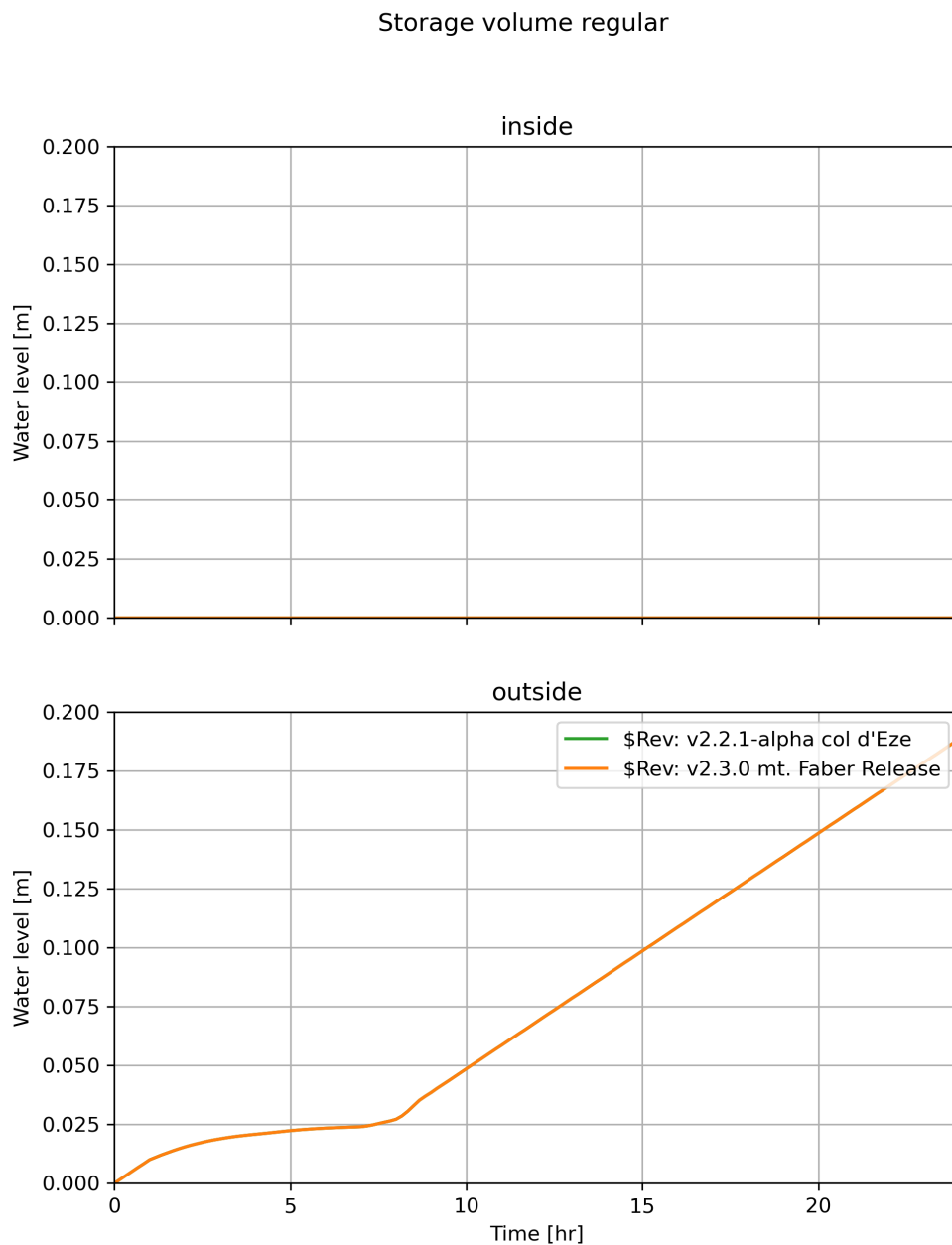


Figure 2.11: Comparison of modelled water level between latest release and stable release.

Table 2.12: Comparison in runtime and modelled water level between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:inside	RMSE loc:outside
\$Rev: v2.2.1-alpha col d'Eze	1.654	-	-
\$Rev: v2.3.0 mt. Faber Release	0.107	0.000	0.000

Experiment description

In this experiment, the implementation of features like storage volume for quadtree subgrid in SFINCS are tested.

Results

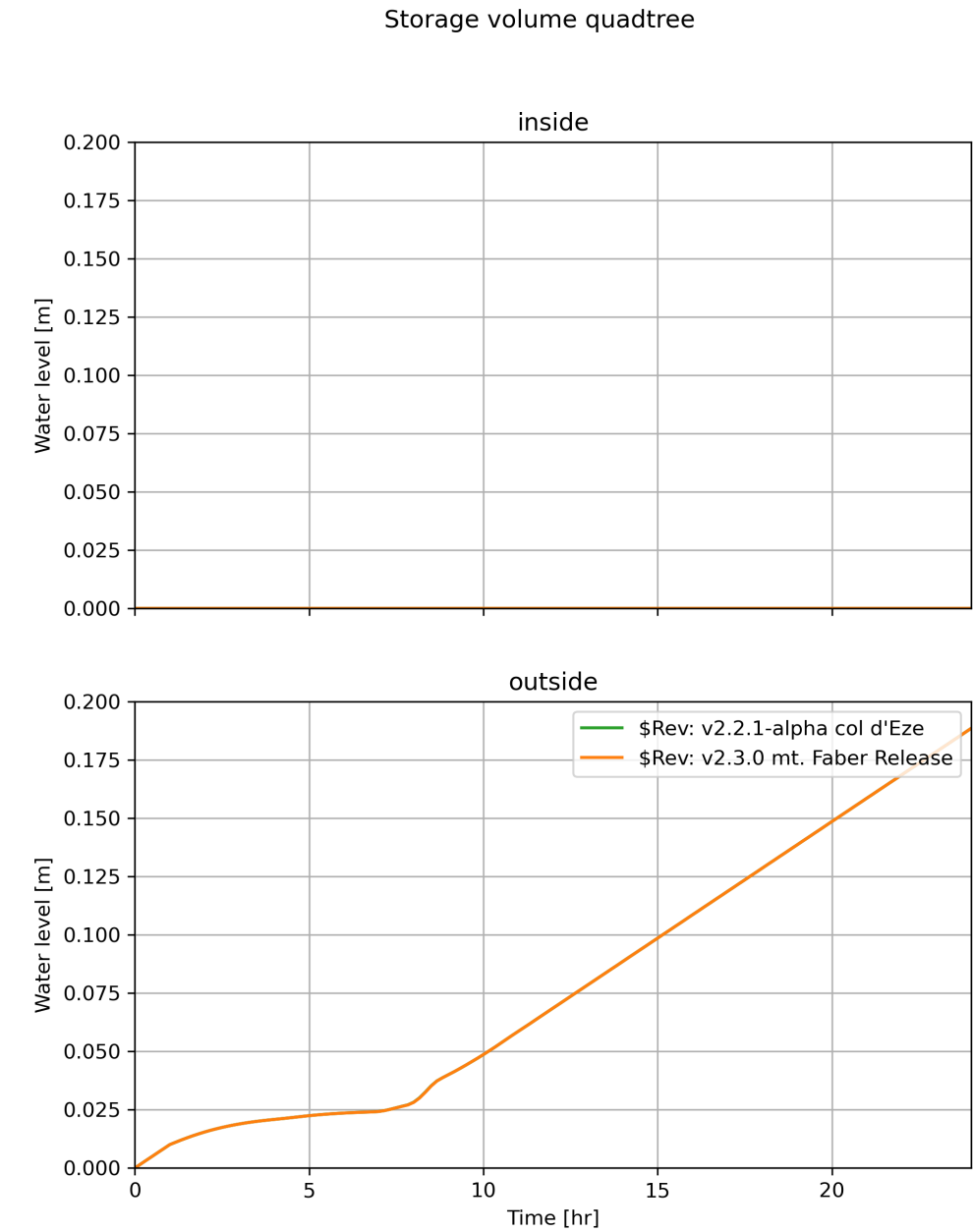


Figure 2.12: Comparison of modelled water level between latest release and stable release.

Table 2.13: Comparison in runtime and modelled water level between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:inside	RMSE loc:outside
\$Rev: v2.2.1-alpha col d'Eze	3.502	-	-
\$Rev: v2.3.0 mt. Faber Release	0.285	0.000	0.000

2.12 Controlled gates

Experiment description

In this experiment, the implementation of the controlled gates feature in SFINCS is tested. The top panel shows the water level outside of the gates, and the lower panel the water level behind the gate. In this experiment, the gate is set to close if the water level exceeds 0.5 m. Also, the gate has a sill at -0.5m, causing the inner water level not to drop below -0.5m,

Results

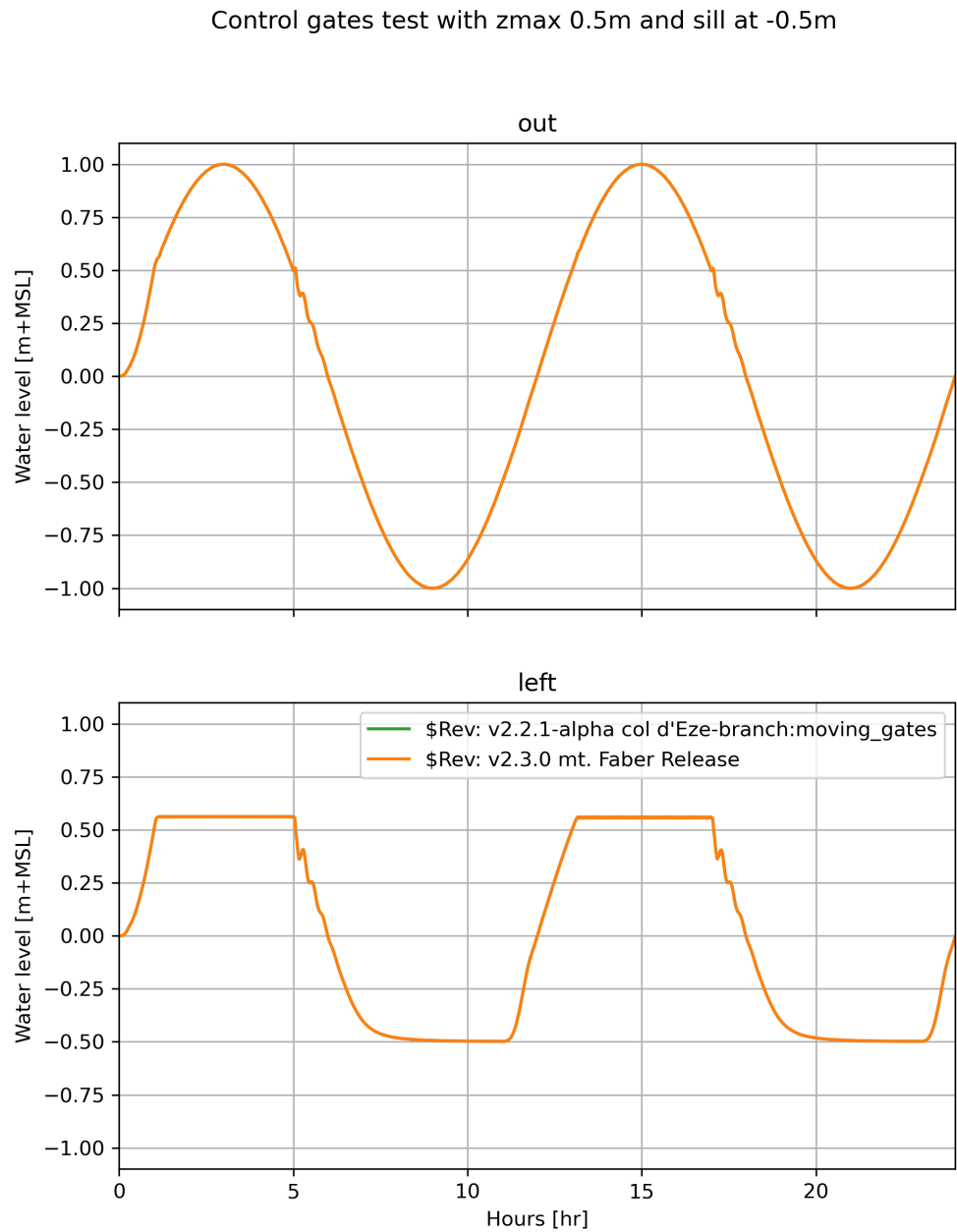


Figure 2.13: Comparison of modelled water levels between latest release and stable release.

Table 2.14: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:out	RMSE loc:left
\$Rev: v2.2.1-alpha col d'Eze-branch:moving_gates	5.281	-	-
\$Rev: v2.3.0 mt. Faber Release	3.808	0.000	0.000

Experiment description

In this experiment, the implementation of the controlled gates feature in SFINCS is tested. The top panel shows the water level outside of the gates, and the lower panel the water level behind the gate. In this experiment, the gate is set to close if the water level exceeds 0.5 m. Also, the gate is set to close if the water level drops below -0.5 m. Due to the closer time of 10 minutes, the final water level inside drops a bit below -0.5m.

Results

Control gates test with $z_{\max}$ 0.5m, $z_{\min}$ at -0.5m and sill at -2.0m

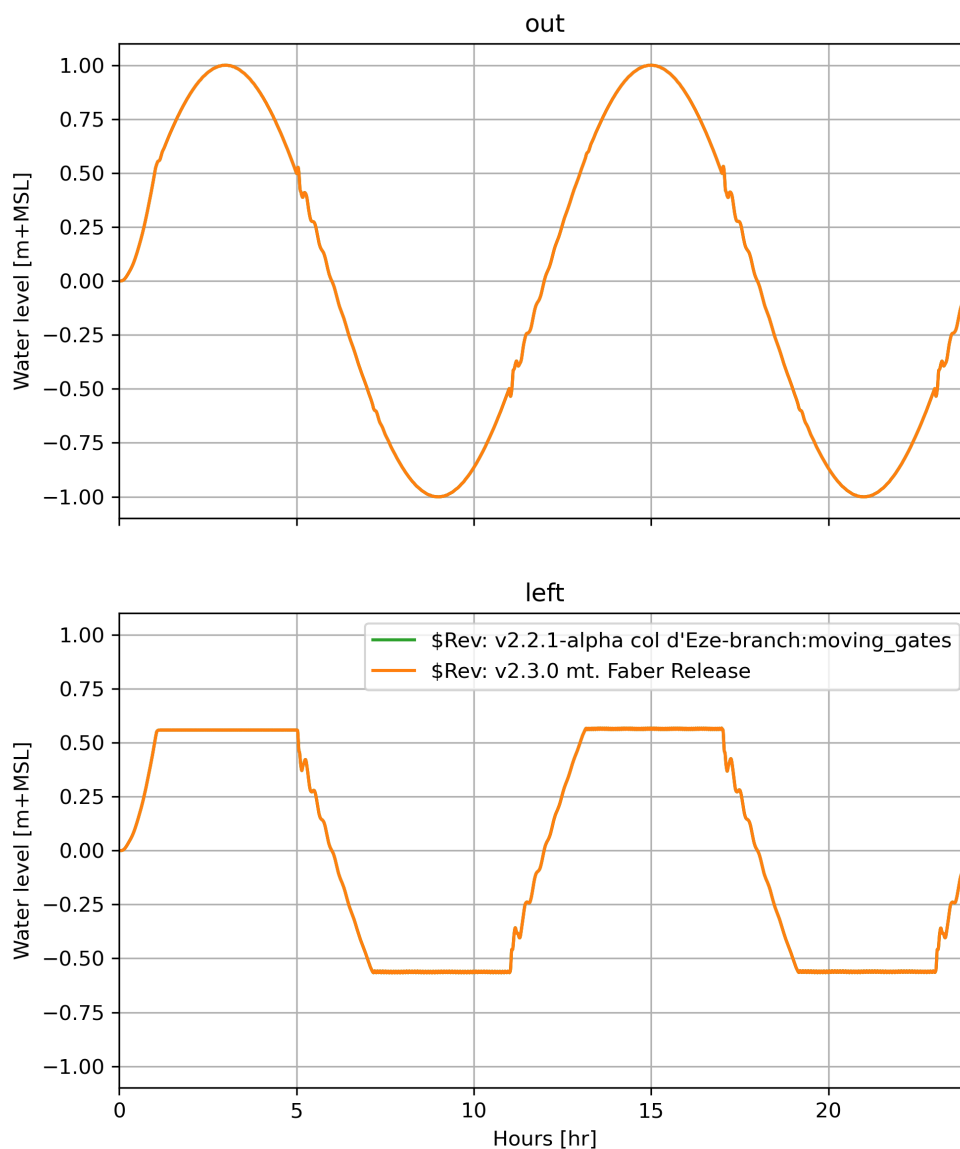


Figure 2.14: Comparison of modelled water levels between latest release and stable release.

Table 2.15: *Comparison in runtime and modelled water levels between latest release and stable release.*

Release version	Total runtime [s]	RMSE loc:out	RMSE loc:left
\$Rev: v2.2.1-alpha col d'Eze-branch:moving_gates	5.229	-	-
\$Rev: v2.3.0 mt. Faber Release	3.152	0.000	0.000

3 Verification

In this chapter, conceptual tests are used to test the performance and numerical implementation of different processes (e.g. flow and waves) in the SFINCS model.

Test	Short description
Bates test	Flow over a horizontal bed
EA test 2	Filling of floodplain depressions
EA test 5	River valley flooding
Dam-break test	Dam-break test with an initial wet bed
Carrier Greenspan	Wave runup
Lijnbak wind growth	Neumann boundary
Globex tests A1, A2	A3

Table 3.1: Overview of the implementation tests in this report.

3.1 Bates test with flow over a horizontal bed

Experiment description

The Bates test demonstrates representation of flow over a horizontal bed, as performed in [Leijnse \(2018\)](#).

Results

A quantitative comparison between the analytical solution and a stable and the latest SFINCS release is presented.

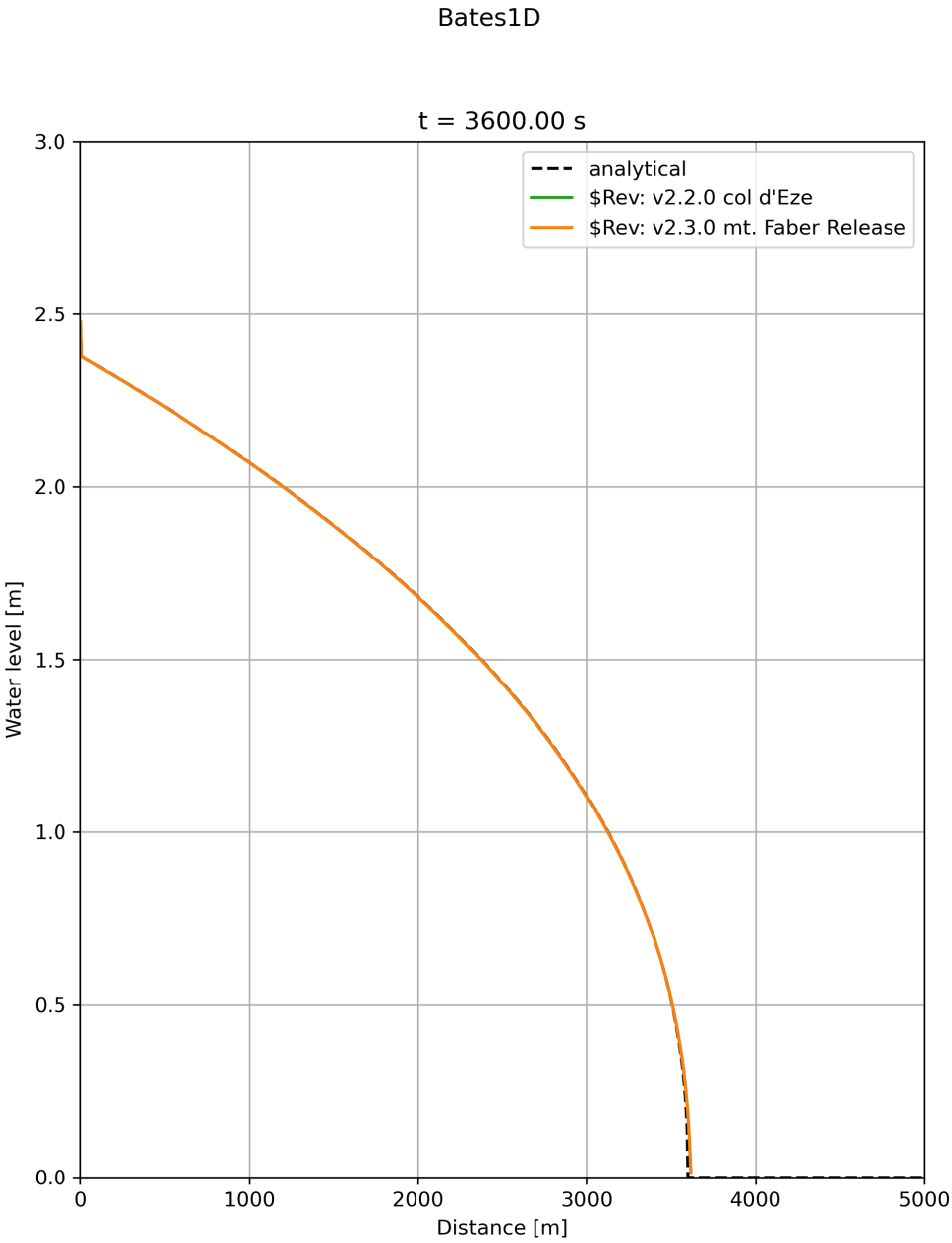


Figure 3.1: Instantaneous water level of the analytical solution and SFINCS (for the stable and latest release) for after 3600 seconds

Table 3.2: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE t=3600.00s
\$Rev: v2.2.0 col d'Eze	0.568	-
\$Rev: v2.3.0 mt. Faber Release	0.188	0.000

3.2 EA test 2: Filling of floodplain depressions

Experiment description

In this test of the Environmental Agency (EA), a flattened egg box-shaped 2000 × 2000 m domain with grid cells of 20 m is simulated. The inflow hydrograph on the top left corner of the domain has a peak flow of 20 m³ /s for 85 min and the total simulation time is 2 days. Water flows into the domain and accumulates in the topographic depressions. The flow depth limiter in the SFINCS model is set to 5 mm and the Manning friction coefficient to $n = 0.03 \text{ s/m}^{1/3}$ (as prescribed by EA). For more details regarding the model setup, we refer to [Néelz and Pender \(2013\)](#) and [Leijnse et al. \(2021\)](#).

Results

A quantitative comparison between a stable and the latest SFINCS release is presented.

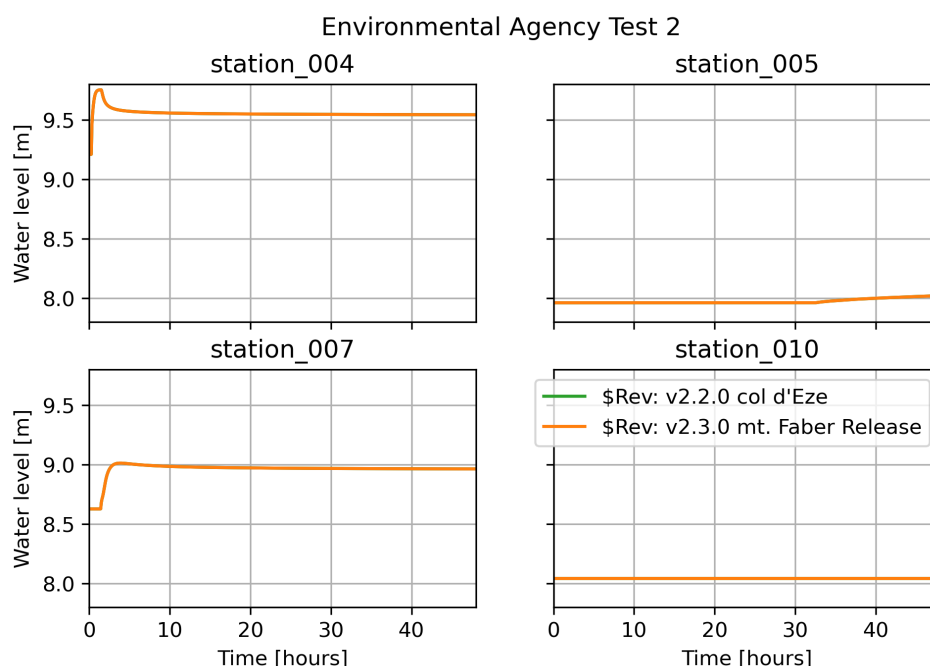


Figure 3.2: Time-series of observed and modeled (with stable and latest SFINCS release) water levels.

Table 3.3: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	4.593	2.612
RMSE loc:station_004	-	0.000
RMSE loc:station_005	-	0.000
RMSE loc:station_007	-	0.000
RMSE loc:station_010	-	0.000

Environmental Agency Test 2

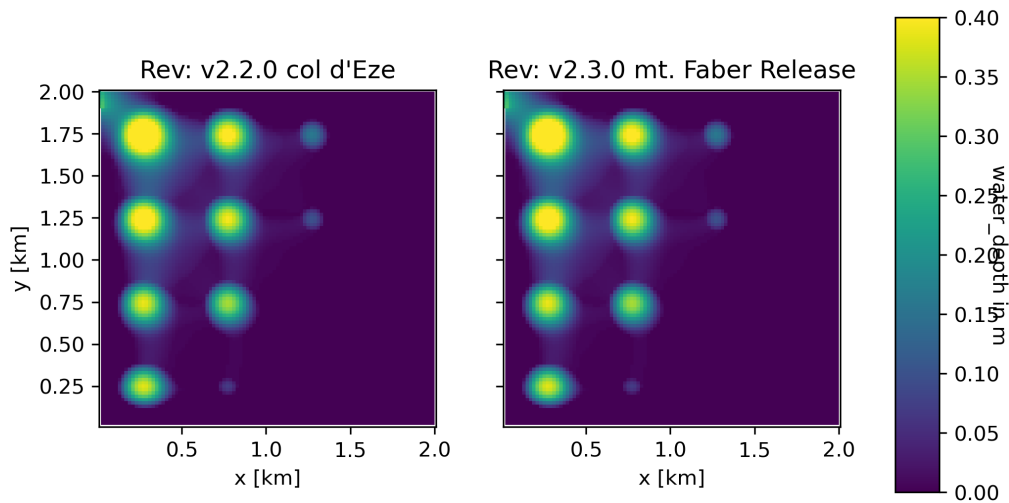


Figure 3.3: Maximum observed and modeled (with stable and latest SFINCS release) water levels.

3.3 EA test 5: River valley flooding

Experiment description

In this test of the Environmental Agency (EA), To verify the ability of the model to simulate flooding in a more realistic case, the river valley flooding test of the benchmark tests of the EA comparison is performed (Test 5). This test is designed to simulate the propagation of a flood wave down a river valley due to a failed dam. A short discharge peak of 3000 m³/s is introduced at the upstream side of the model in the first 100 min of the simulation, where the total simulation time is 30 h. The grid size is 50 m and the SFINCS model contains 7640 active points. The flood wave propagates through the domain due to a total head difference of 60 m from up-to downstream. The flow depth limiter in the SFINCS model is set to 5 mm and the Manning friction coefficient to $n = 0.04 \text{ s/m}^{1/3}$ (as prescribed by EA). For more details regarding the model setup, see [Néelz and Pender \(2013\)](#) and [Leijnse *et al.* \(2021\)](#).

Results

A quantitative comparison between a stable and the latest SFINCS release is presented.

Environmental Agency Test 5

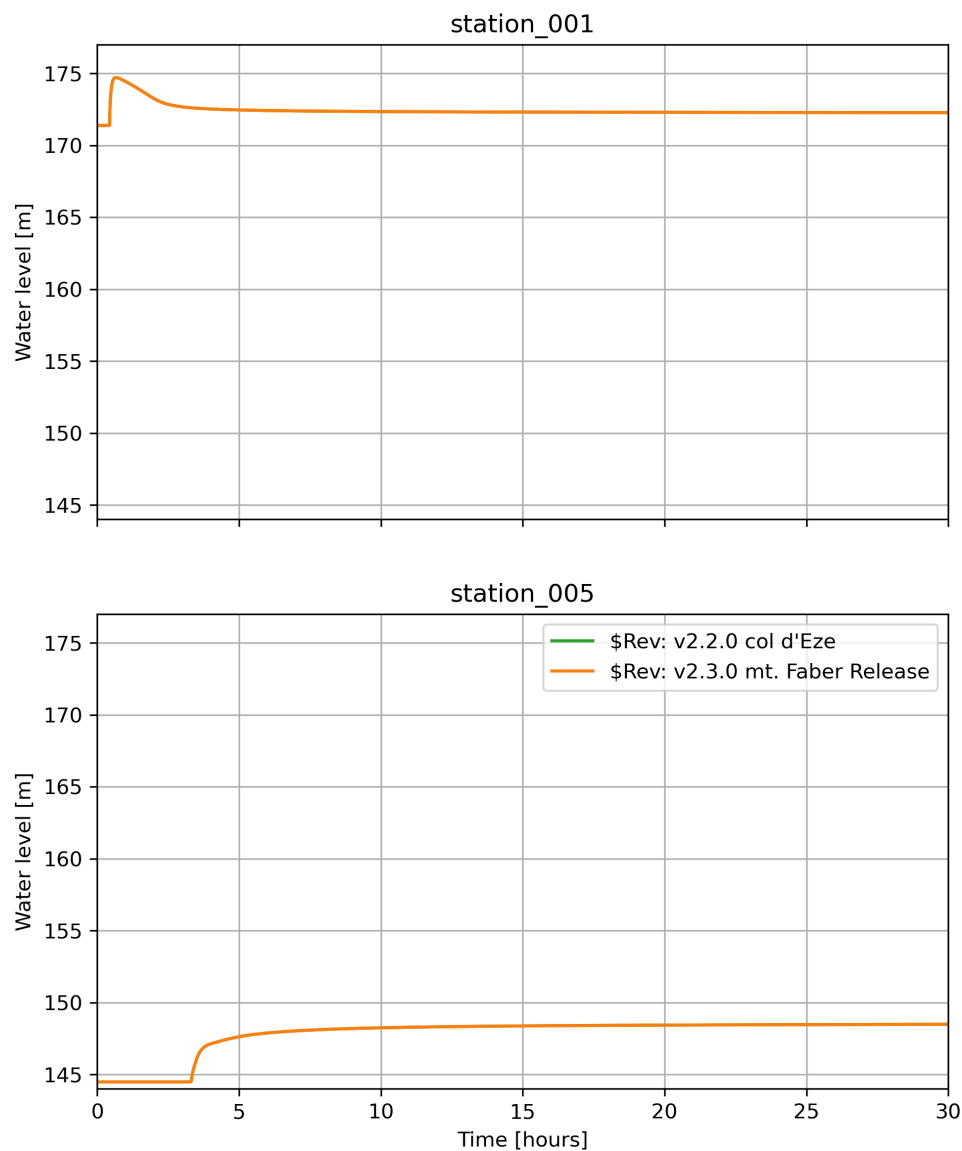


Figure 3.4: Time-series of observed and modeled (with stable and latest SFINCS release) water levels.

Table 3.4: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:station_001	RMSE loc:station_005
\$Rev: v2.2.0 col d'Eze	4.371	-	-
\$Rev: v2.3.0 mt. Faber Release	2.338	0.000	0.000

Environmental Agency Test 5

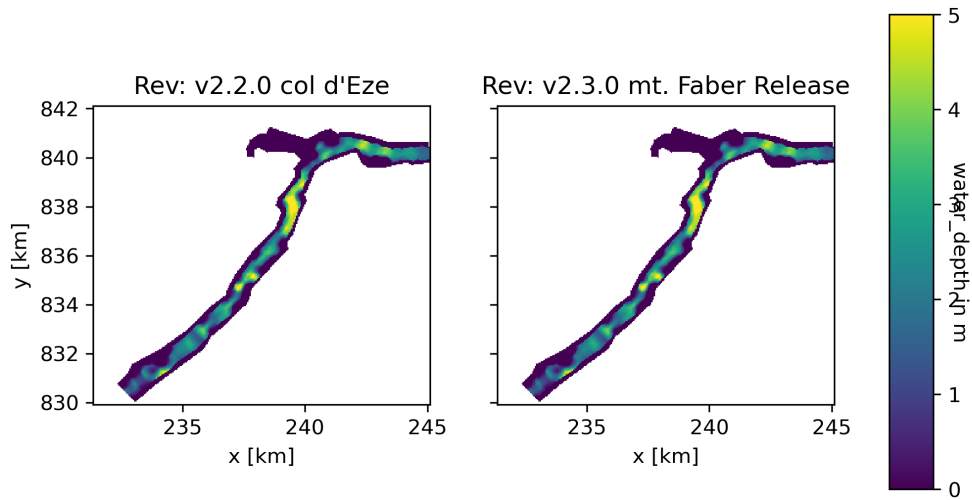


Figure 3.5: Maximum observed and modeled (with stable and latest SFINCS release) water levels.

3.4 Dam-break test with an initial wet bed

Experiment description

The dam break test with an initial wet bed demonstrates the improvement in accuracy by adding the advection term to model supercritical flow. The setup is based on [Stelling and Duinmeijer \(2003\)](#) and [Cui \(2013\)](#) with the analytical solution as in [Chanson \(2006\)](#). The water depth is 1 m upstream of the dam break and 0.1 m downstream. The model is 100 m long with a grid size of 0.1 m. In order to compare to the analytical solution SFINCS is run without friction, see: [Leijnse et al. \(2021\)](#).

Results

Without the advection term (SFINCS-LIE), the model results differ greatly from the analytical solution (not shown here). The forward propagating speed is underestimated while the water depth at the front of the shock wave is overestimated. When the advection term (SFINCS-SSWE) is included the agreement with the analytical solution is drastically improved. A quantitative comparison between the analytical solution and a stable and the latest SFINCS release is presented.

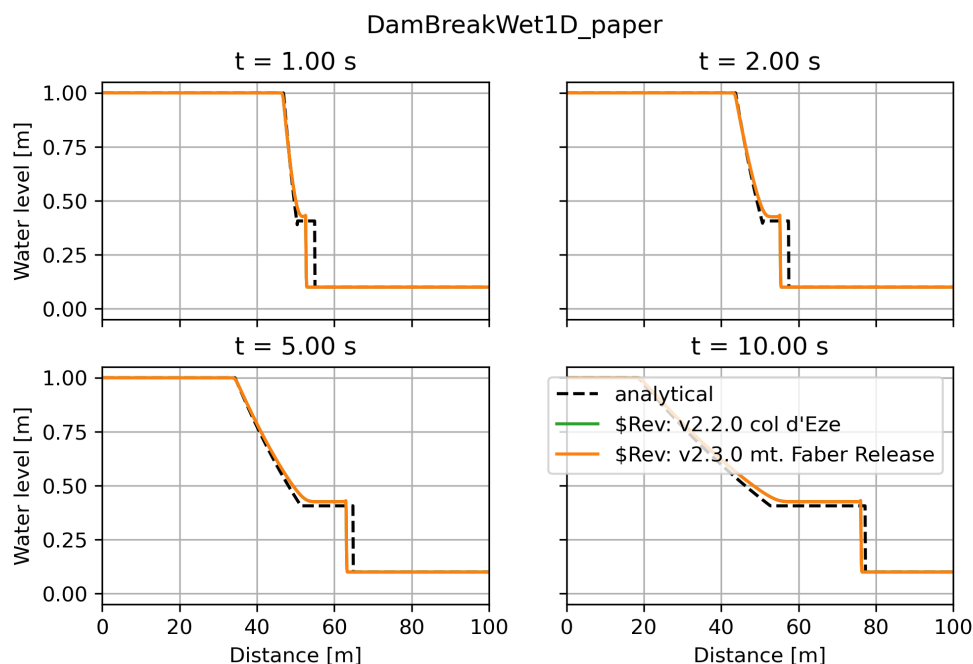


Figure 3.6: Instantaneous water level of the analytical solution and SFINCS with advection (for the stable and latest release) for the dam break test with wet bed after 1 (upper-left), 2 (upper-right), 5 (lower-left) and 10 (lower-right) seconds

Table 3.5: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	0.102	0.114
RMSE t=1.00s	-	0.000
RMSE t=2.00s	-	0.000
RMSE t=5.00s	-	0.000
RMSE t=10.00s	-	0.000

3.5 Carrier Greenspan (1958) wave runup

Experiment description

To verify the model performance to simulate wave runup, model results of SFINCS are compared to the analytical solution derived by [Carrier and Greenspan \(1958\)](#) for runup and run-down of non-breaking long waves on a plane beach. A free long wave with a wave period of 32 s and wave amplitude of half the wave breaking amplitude propagates over a beach with a constant slope equal to 1:25 (as in [Van Dongeren and Svendsen \(1997\)](#)). The model consists of a 150 m long domain and grid sizes of 0.05 m. It simulates 6.5 min and is forced with a sinusoidal signal at the seaward boundary. The flow depth limiter is set to 0.0001 m and, like in the analytical solution, no friction is used, see: [Leijnse et al. \(2021\)](#).

Results - without advection

A quantitative comparison between a stable and the latest SFINCS release without advection is presented.

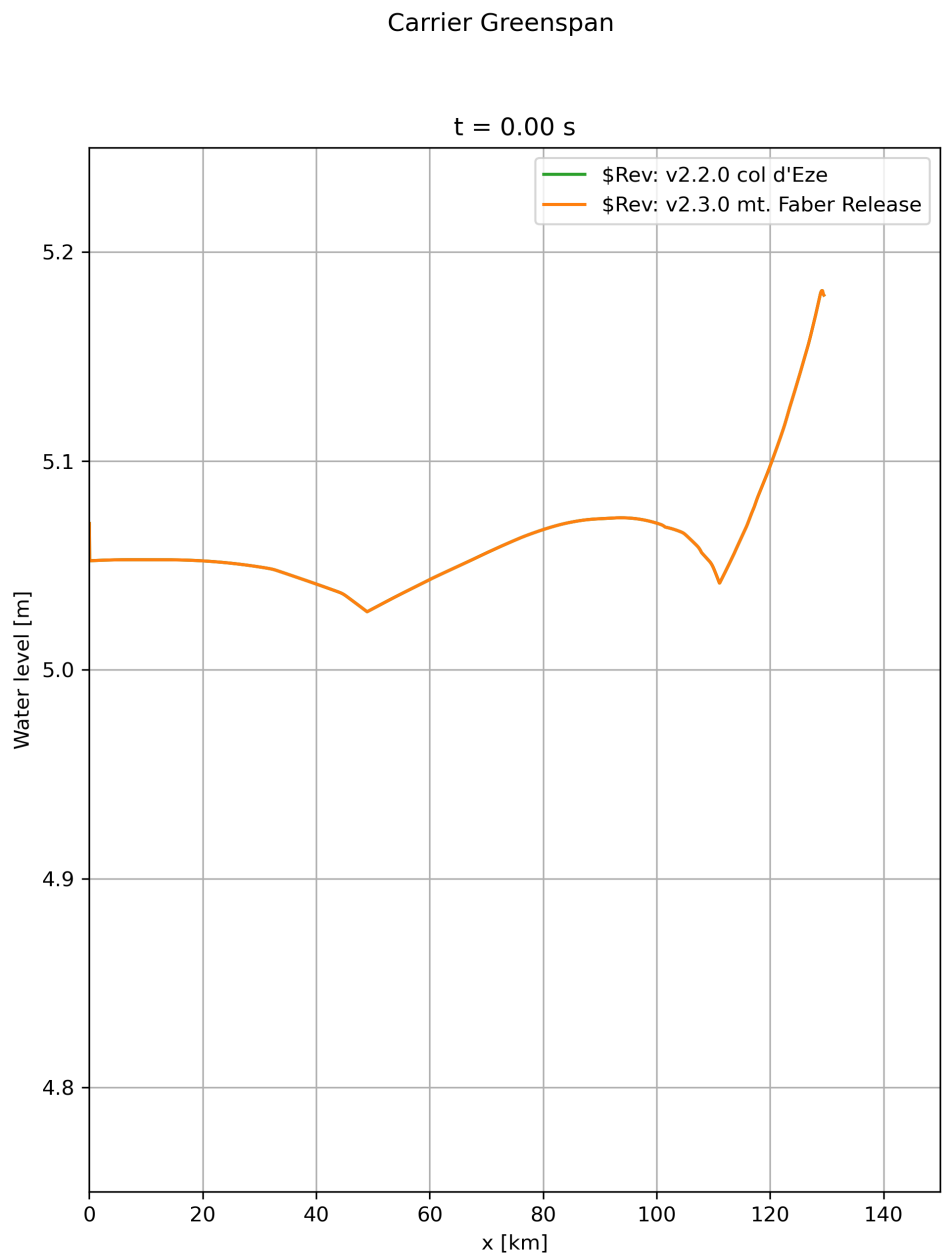


Figure 3.7: Maximum observed and modeled (with stable and latest SFINCS release without advection) water levels.

Table 3.6: Comparison in runtime and modelled water levels between latest release and stable release without advection.

Release version	Total runtime [s]	RMSE t=0.00s
\$Rev: v2.2.0 col d'Eze	6.701	-
\$Rev: v2.3.0 mt. Faber Release	13.031	0.000

Results - with advection

A quantitative comparison between a stable and the latest SFINCS release is presented.

Carrier Greenspan - advection

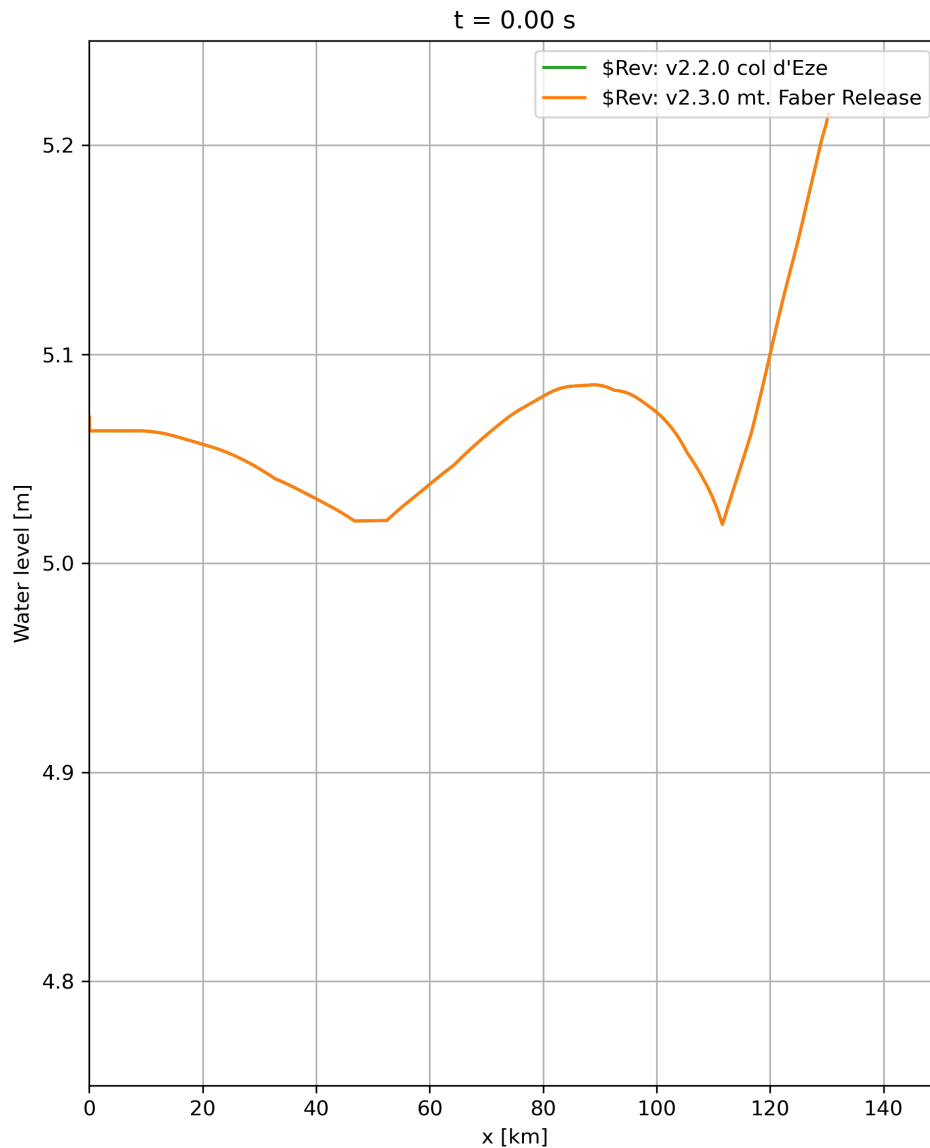


Figure 3.8: Observed and modeled (with stable and latest SFINCS release) water levels.

Table 3.7: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE t=0.00s
\$Rev: v2.2.0 col d'Eze	7.674	-
\$Rev: v2.3.0 mt. Faber Release	3.896	0.000

4 Application

In this chapter, case studies are performed and discussed in order to assess whether the conclusions of the conceptual tests from [chapter 3](#) also apply in more realistic cases.

Test	Short description
Jacksonville during Hurricane Irma (2017)	Compound flooding case
Hernani during Haiyan (2013)	Wave driven flooding case
Houston during Hurricane Harvey (2017) including netcdf sbgfile	Rainfall driven flooding
Charleston during Hurricane Hugo (1989)	Storm surge flooding
Bahamas quadtree model	synthetic tropical cyclone model
St. Johns river tide	Tidal propagation up the St. Johns river
Riverine flooding	Puget Sound flooding event 2009 and 2020
Large scale spherical TC model	netspwfile input
Internal tide generation	with and without superimposed surge
Duck	SnapWave IG model, with and without rising water level
Outer Banks	SFINCS quadtree SnapWave IG model
SFINCS release movie	Flooded embedded SFINCS logo
SFINCS release movie 2023 version	Flooded embedded SFINCS logo

Table 4.1: Overview of the implementation tests in this report.

4.1 Jacksonville during Hurricane Irma (2017)

Experiment description

Hurricane Irma (2017) caused a major compound flooding event in Jacksonville (Florida, USA) due to a combination of fluvial, pluvial, tidal and wind-driven processes, the latter being a local wind and a large-scale surge component. Floodwaters rushed into the city's streets and reached up to 1.5 m deep in some locations ([Cangialosi et al. \(2018\)](#)). A model chain with different nested models is used to hindcast the hydrodynamic response to this hurricane for the Jacksonville area, see: [Leijnse et al. \(2021\)](#).

Results - without advection

To assess the performance of the models without advection, computed water levels are compared to observed time-series at observation stations of the USGS (USGS, 2017) and NOAA (NOAA, 2017) at a number of locations inland and along the rivers of the considered domain. Computed water levels close to the coast are underestimated compared to observations (sta-

tions Fernandina Beach, Mayport, FLSTJ03118 and FLSTJ03125) for all three models. The water levels in the region near the city of Jacksonville where the river narrows (stations Main Street Bridge and I-295 Bridge in the St. Johns river) are correctly modeled by all three models. Further south along the river, the water levels are underestimated again. In the two stations in the St. Mary's River (GACAM17830 and GACAM17823) this difference between the models and the data does not occur, though only the peaks are measured in the observation data (no continuous observations present).

Jacksonville

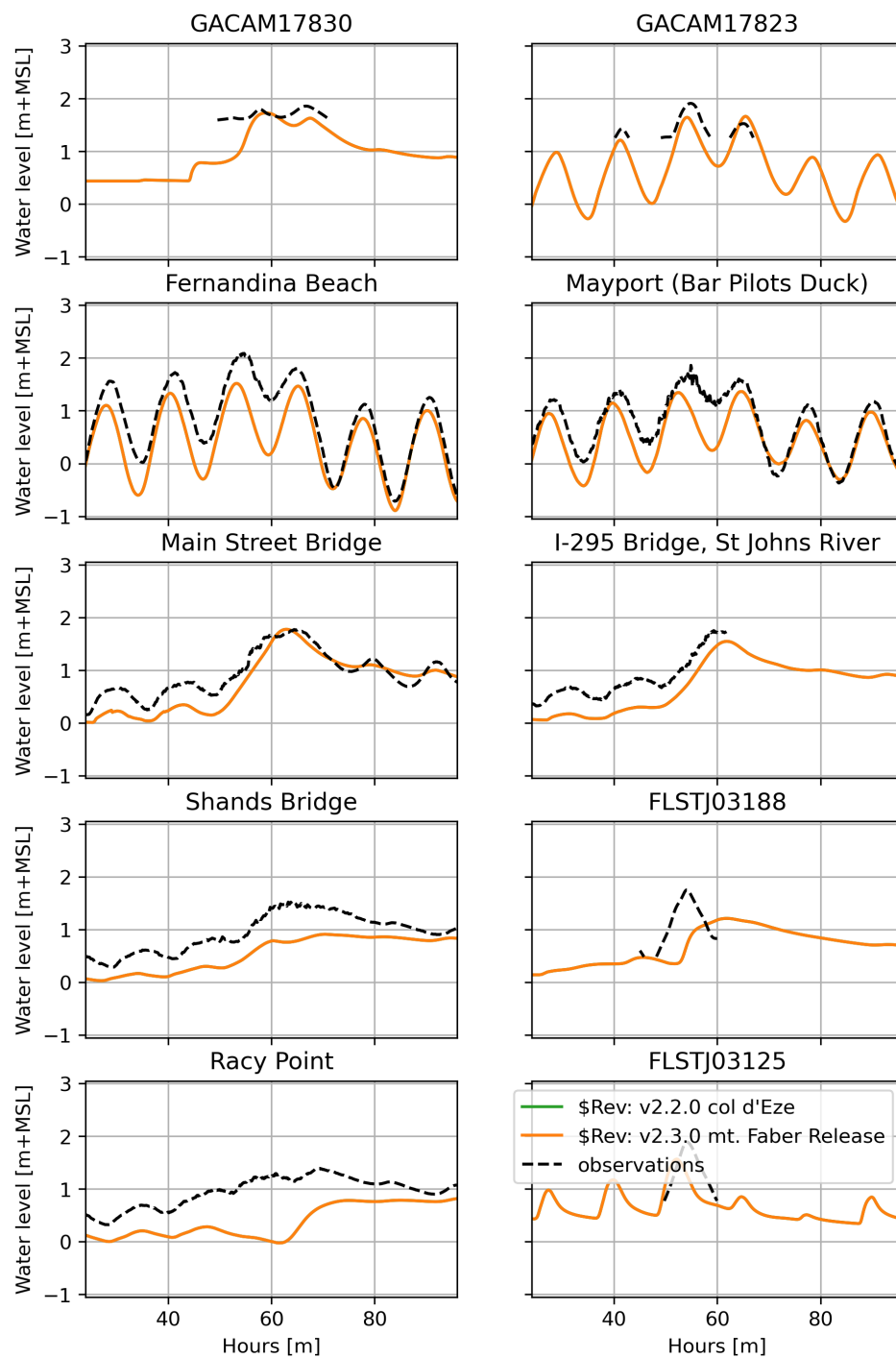


Figure 4.1: Time-series of observed and modeled (with stable and latest SFINCS release without advection) water levels during Hurricane Irma.

Table 4.2: Comparison in runtime and modelled water levels between latest release and stable release without advection.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	1223.982	1232.788
RMSE loc:GACAM17830	-	0.000
RMSE loc:GACAM17823	-	0.000
RMSE loc:Fernandina Beach	-	0.001
RMSE loc:Mayport (Bar Pilots Duck)	-	0.000
RMSE loc:Main Street Bridge	-	0.000
RMSE loc:I-295 Bridge, St Johns River	-	0.000
RMSE loc:Shands Bridge	-	0.000
RMSE loc:FLSTJ03188	-	0.000
RMSE loc:Racy Point	-	0.000
RMSE loc:FLSTJ03125	-	0.000

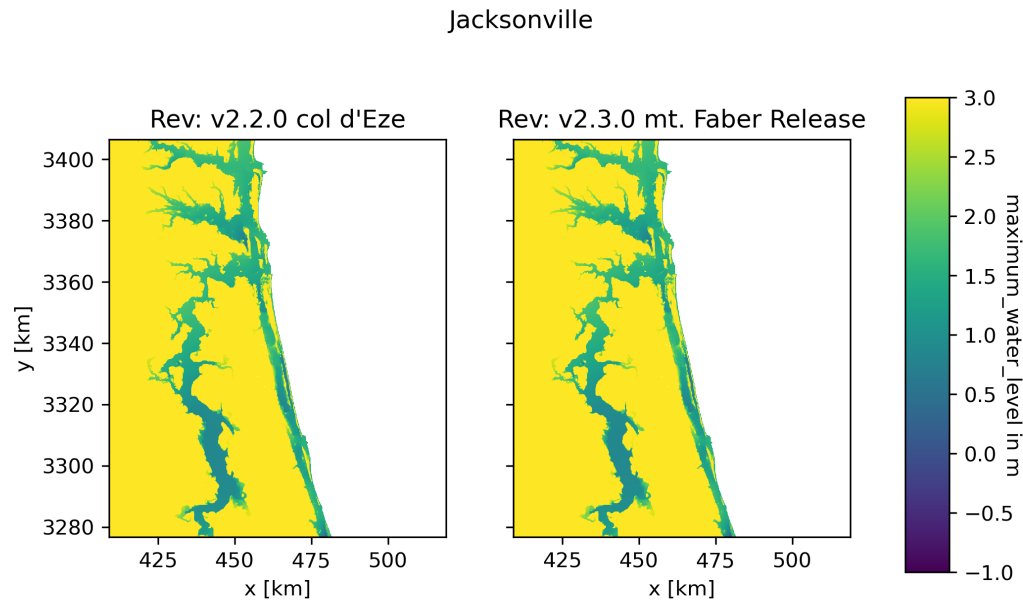


Figure 4.2: Maximum observed and modeled (with stable and latest SFINCS release without advection) water levels during Hurricane Irma.

Results - with advection

To assess the performance of the models including advection, computed water levels are compared to observed time-series at observation stations of the USGS (USGS, 2017) and NOAA (NOAA, 2017) at a number of locations inland and along the rivers of the considered domain.

Jacksonville

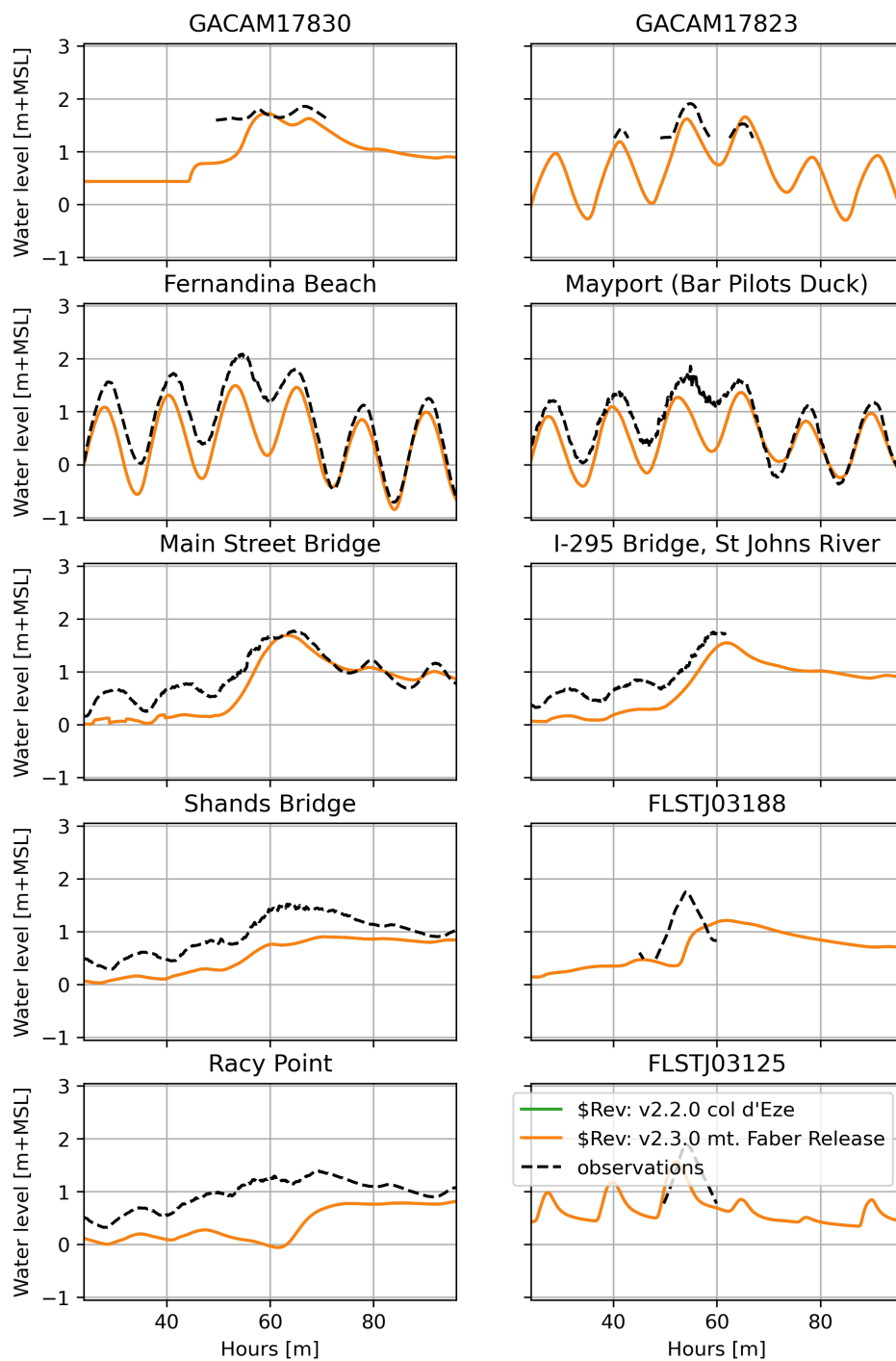


Figure 4.3: Time-series of observed and modeled (with stable and latest SFINCS release including advection) water levels during Hurricane Irma.

Table 4.3: Comparison in runtime and modelled water levels between latest release and stable release including advection.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	1423.423	1362.835
RMSE loc:GACAM17830	-	0.000
RMSE loc:GACAM17823	-	0.000
RMSE loc:Fernandina Beach	-	0.000
RMSE loc:Mayport (Bar Pilots Duck)	-	0.000
RMSE loc:Main Street Bridge	-	0.000
RMSE loc:I-295 Bridge, St Johns River	-	0.000
RMSE loc:Shands Bridge	-	0.000
RMSE loc:FLSTJ03188	-	0.000
RMSE loc:Racy Point	-	0.000
RMSE loc:FLSTJ03125	-	0.000

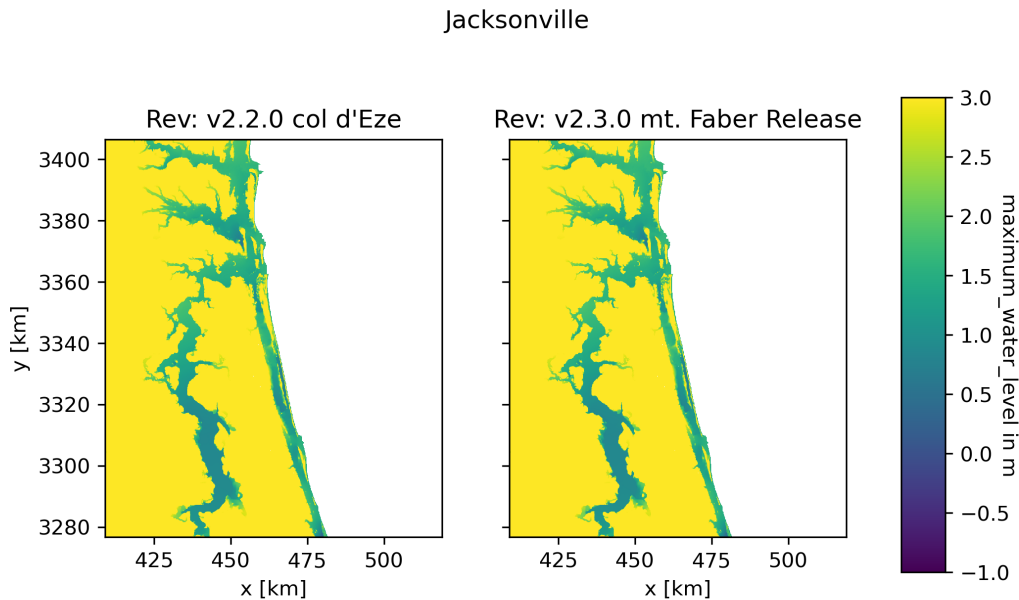


Figure 4.4: Maximum observed and modeled (with stable and latest SFINCS release including advection) water levels during Hurricane Irma.

4.2 Hernani during Haiyan (2013)

Experiment description

Wave-driven flooding event that occurred in 2013, when Typhoon Haiyan caused major damage in the Philippines with a death toll of over 6300 and millions of people affected (Reduction and Council) (2013). In this case for the village of Hernani, Eastern Samar, large infragravity waves were generated over the fringing reef in front of the village, sweeping away whole houses despite the sea wall in front of the village (see pictures in Roeber and Bricker (2015)).

Results - without advection

To assess the performance of the models without advection, computed water levels are

compared to modelled waterlevel by the advanced wave resolving XBeach non-hydrostatic+ model, see: [Leijnse et al. \(2021\)](#).

Hernani

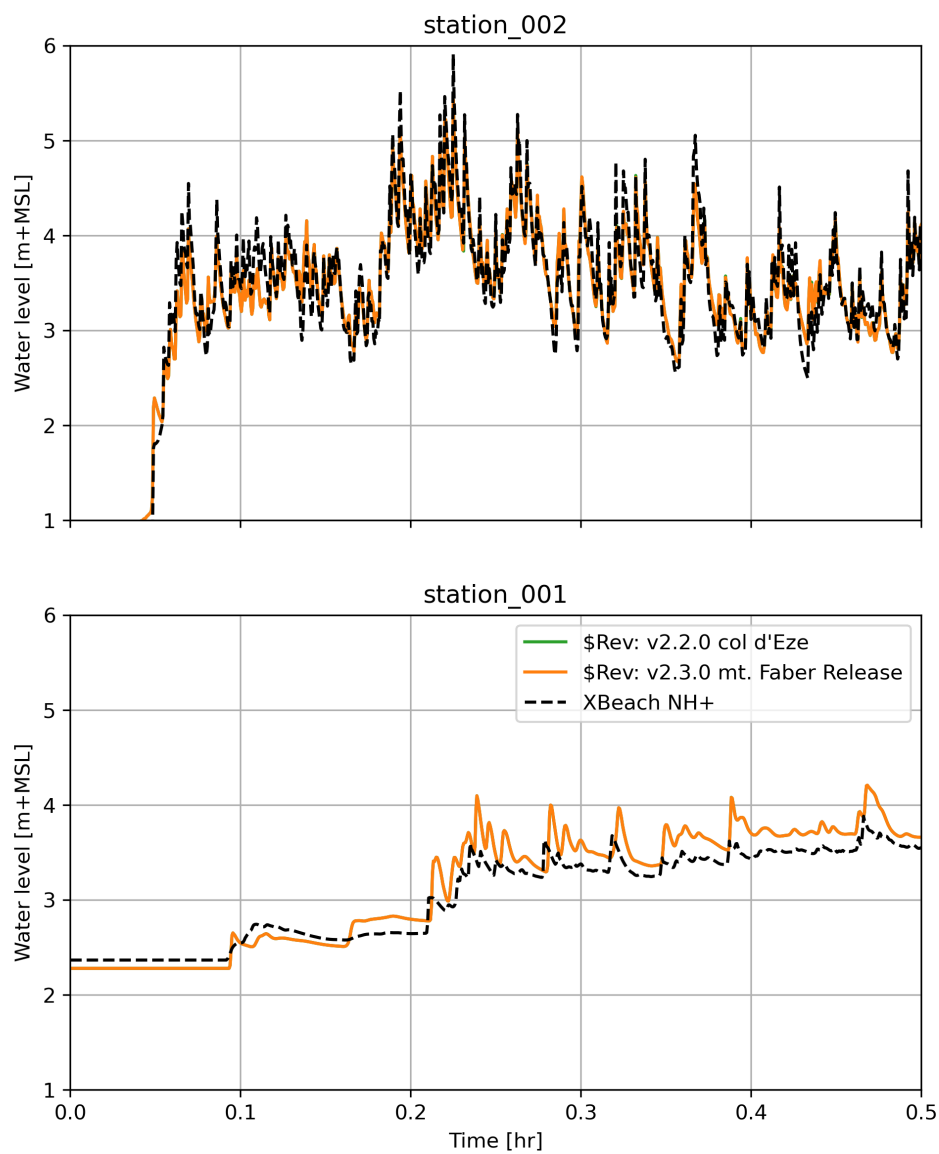


Figure 4.5: Time-series of observed and modeled (with stable and latest SFINCS release without advection) water levels during Hurricane Irma.

Table 4.4: Comparison in runtime and modelled water levels between latest release and stable release without advection.

Release version	Total runtime [s]	RMSE loc:station_002	RMSE loc:station_001
\$Rev: v2.2.0 col d'Eze	5.779	-	-
\$Rev: v2.3.0 mt. Faber Release	7.191	0.011	0.001

Hernani

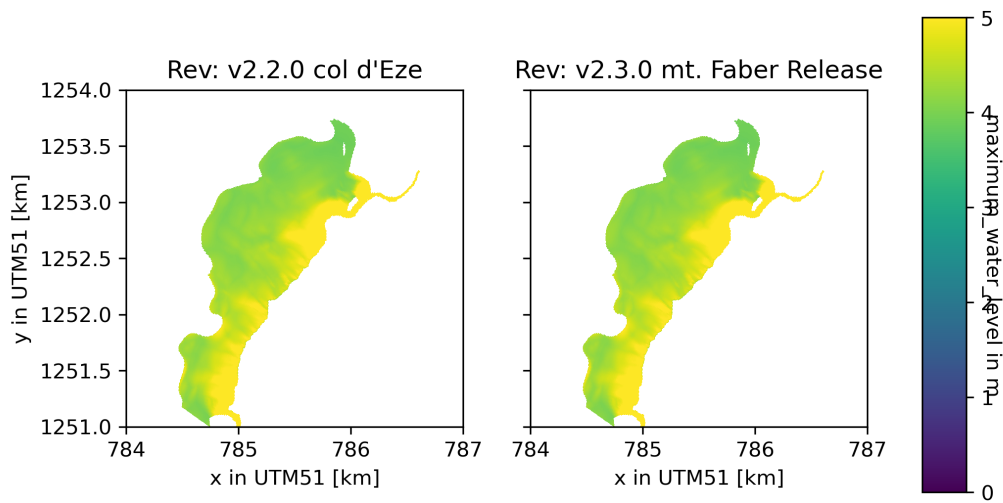


Figure 4.6: Maximum observed and modeled (with stable and latest SFINCS release without advection) water levels during Hurricane Irma.

Results - with advection

To assess the performance of the models with advection, computed water levels are compared to modelled waterlevel by the advanced wave resolving XBeach non-hydrostatic+ model.

Hernani with advection

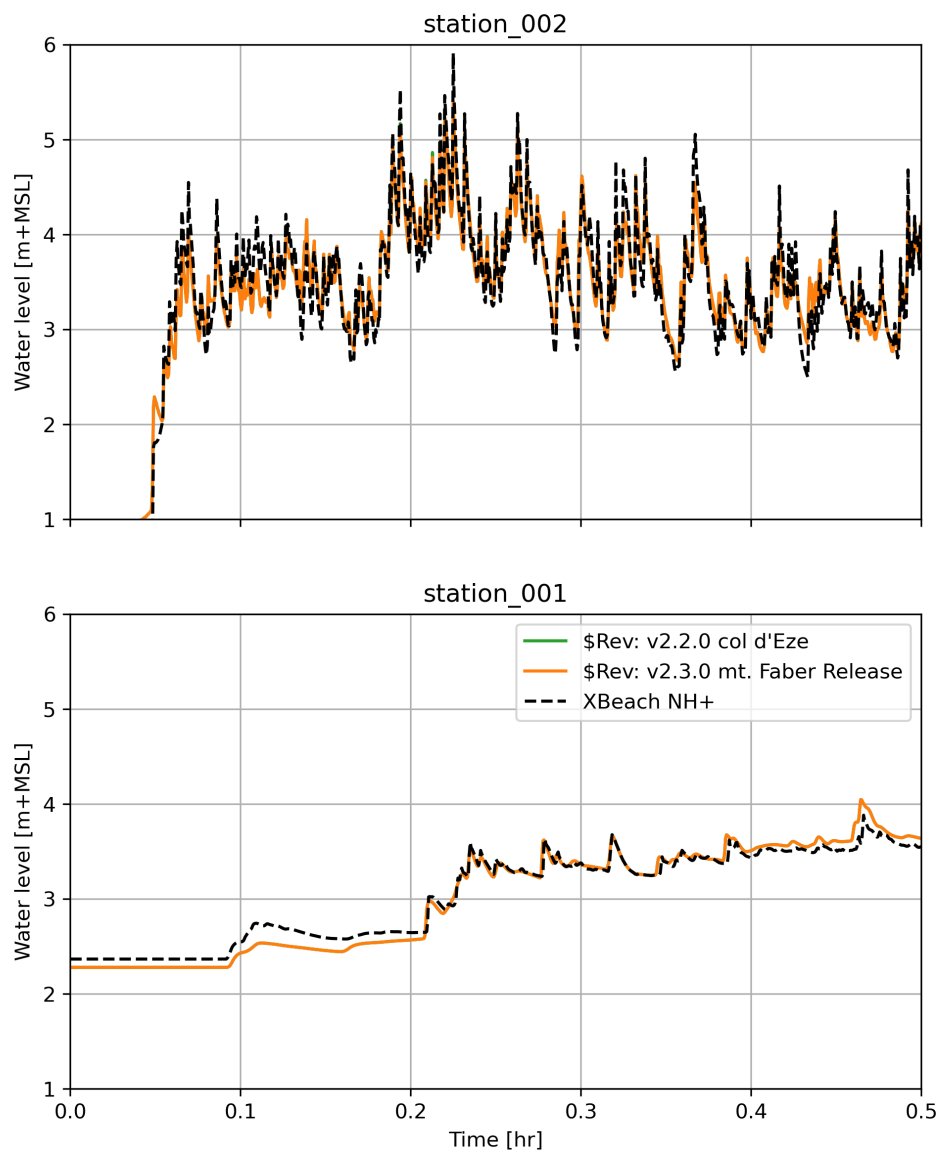


Figure 4.7: Time-series of observed and modeled (with stable and latest SFINCS release with advection) water levels during Hurricane Irma.

Table 4.5: Comparison in runtime and modelled water levels between latest release and stable release with advection.

Release version	Total runtime [s]	RMSE loc:station_002	RMSE loc:station_001
\$Rev: v2.2.0 col d'Eze	7.175	-	-
\$Rev: v2.3.0 mt. Faber Release	2.330	0.008	0.001

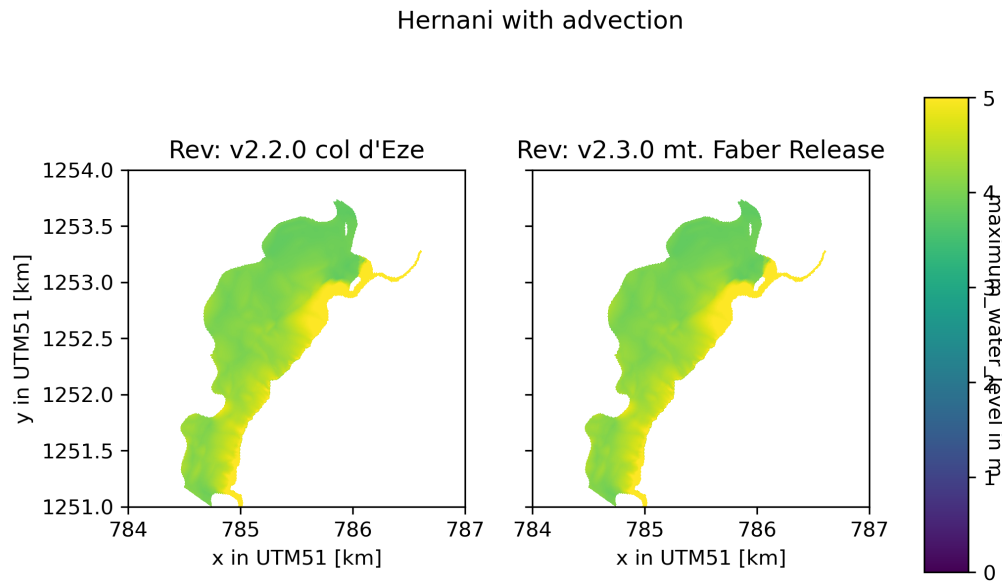


Figure 4.8: Maximum observed and modeled (with stable and latest SFINCS release with advection) water levels during Hurricane Irma.

4.3 Houston during Hurricane Harvey (2017)

Experiment description

Hurricane Harvey was a devastating Category 4 hurricane that made landfall on Texas and Louisiana in August 2017, causing catastrophic flooding and more than 100 deaths. It is tied with 2005's Hurricane Katrina as the costliest tropical cyclone on record inflicting 125 billion (2017 USD) in damage, primarily from catastrophic rainfall-triggered flooding in the Houston metropolitan area.

[Sebastian *et al.* \(2021\)](#) demonstrated the ability of the SFINCS model to delineate the depth and extent of flooding during Hurricane Harvey in Houston, Texas. The model was validated against water level time-series at twenty-one United States Geological Survey (USGS) observation points and 115 high watermark locations. However, in this testbed, we assess the model's performance to 6 observed time series across the city with a 500-meter subgrid version of SFINCS. In particular, the model can reproduce the observed rise and fall of water levels at observations 5, 14, 20, 22 and 24. However, the model does not compute enough of an increase in water levels at observation 9.

Results

To assess the performance of the models, computed water levels are compared to observed time-series at observation stations along the rivers and bayous of Houston.

Hurricane Harvey (2017)

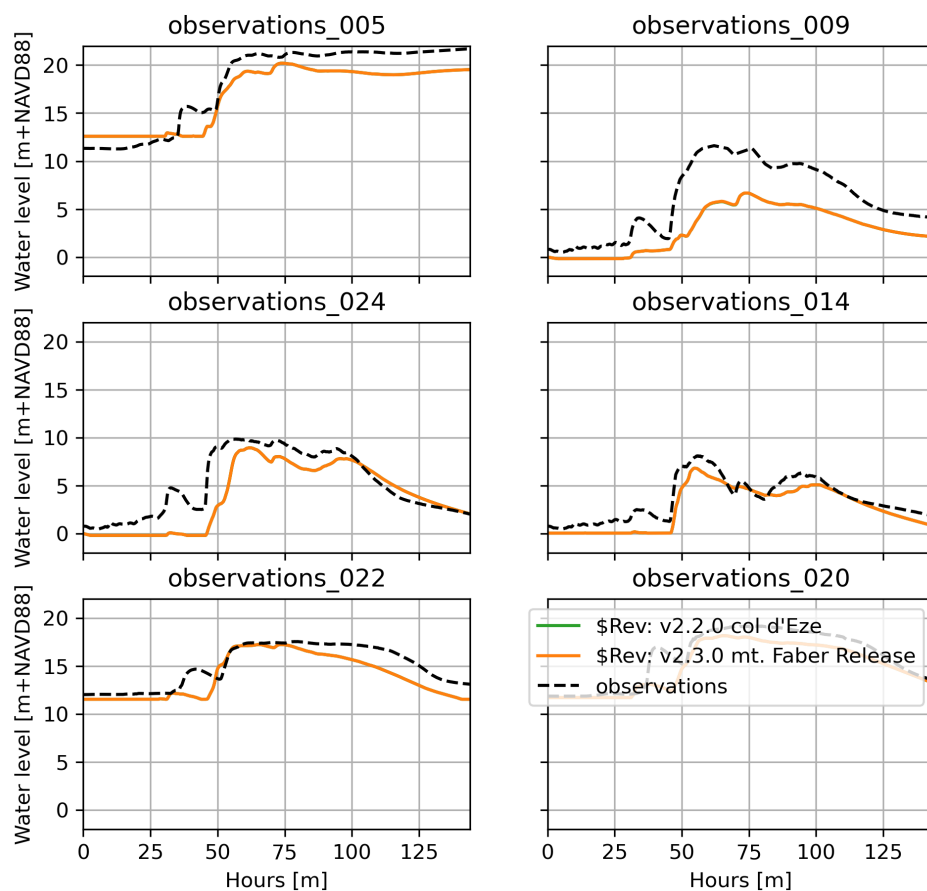


Figure 4.9: Time-series of observed and modeled (with stable and latest SFINCS release) water levels during Hurricane Harvey.

Table 4.6: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	7.869	4.072
RMSE loc:observations_005	-	0.000
RMSE loc:observations_009	-	0.010
RMSE loc:observations_024	-	0.000
RMSE loc:observations_014	-	0.000
RMSE loc:observations_022	-	0.000
RMSE loc:observations_020	-	0.000

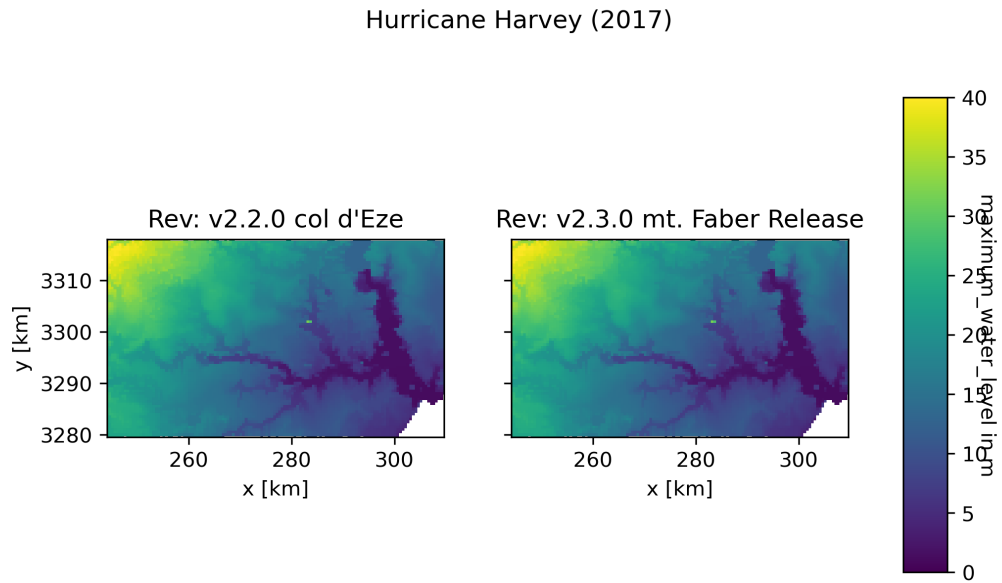


Figure 4.10: Maximum observed and modeled (with stable and latest SFINCS release) water levels during Hurricane Harvey.

Experiment description

500 meter resolution version of the Harvey model with new netcdf subgrid input as in Van Ormondt et al. (2024).

Results

To assess the performance of the models, computed water levels are compared to observed time-series at observation stations along the rivers and bayous of Houston.

Hurricane Harvey (2017)

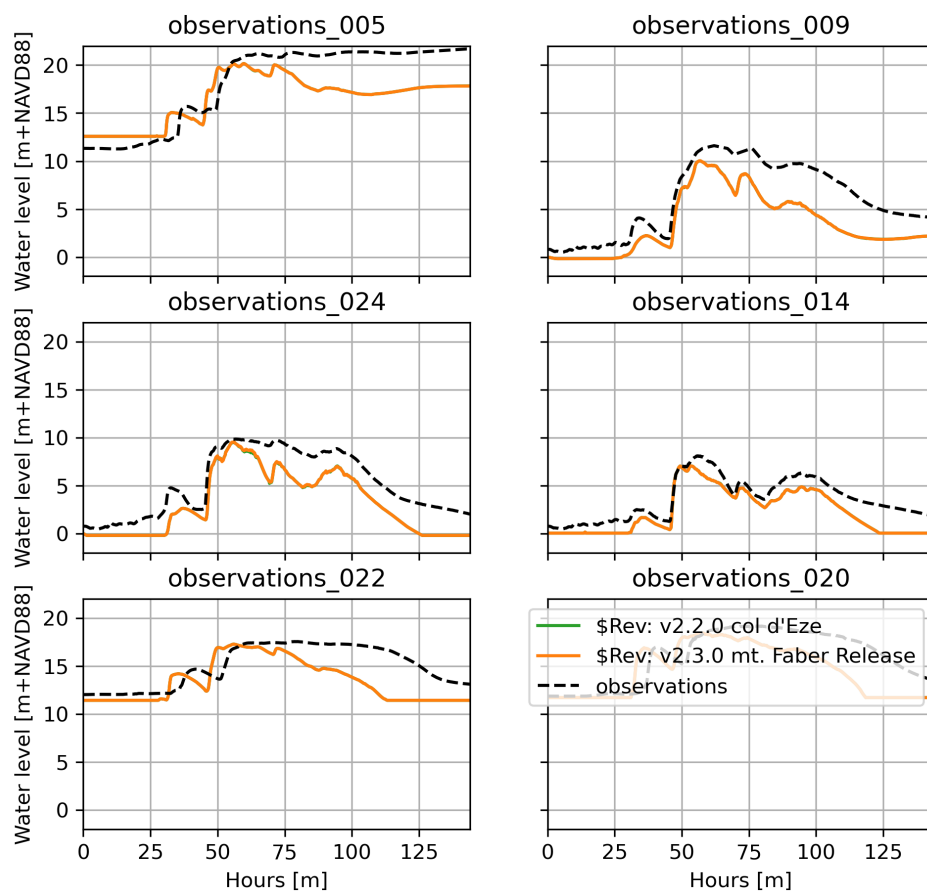


Figure 4.11: Time-series of observed and modeled (with stable and latest SFINCS release) water levels during Hurricane Harvey.

Table 4.7: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	8.468	16.728
RMSE loc:observations_005	-	0.006
RMSE loc:observations_009	-	0.011
RMSE loc:observations_024	-	0.035
RMSE loc:observations_014	-	0.007
RMSE loc:observations_022	-	0.001
RMSE loc:observations_020	-	0.002

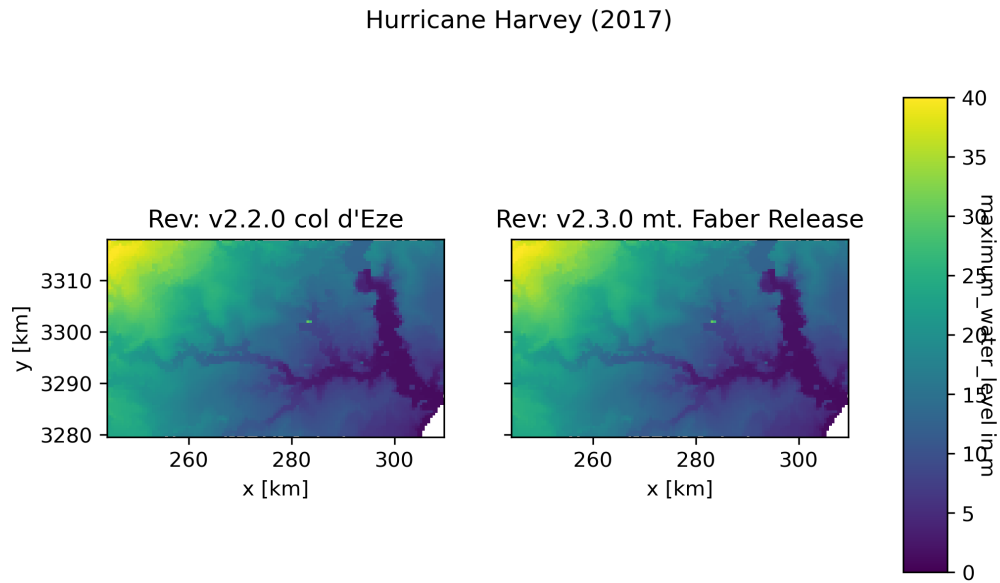


Figure 4.12: Maximum observed and modeled (with stable and latest SFINCS release) water levels during Hurricane Harvey.

Experiment description

100 meter resolution version of the Harvey model with new netcdf subgrid input as in Van Ormondt et al. (2024).

Results

To assess the performance of the models, computed water levels are compared to observed time-series at observation stations along the rivers and bayous of Houston.

Hurricane Harvey (2017)

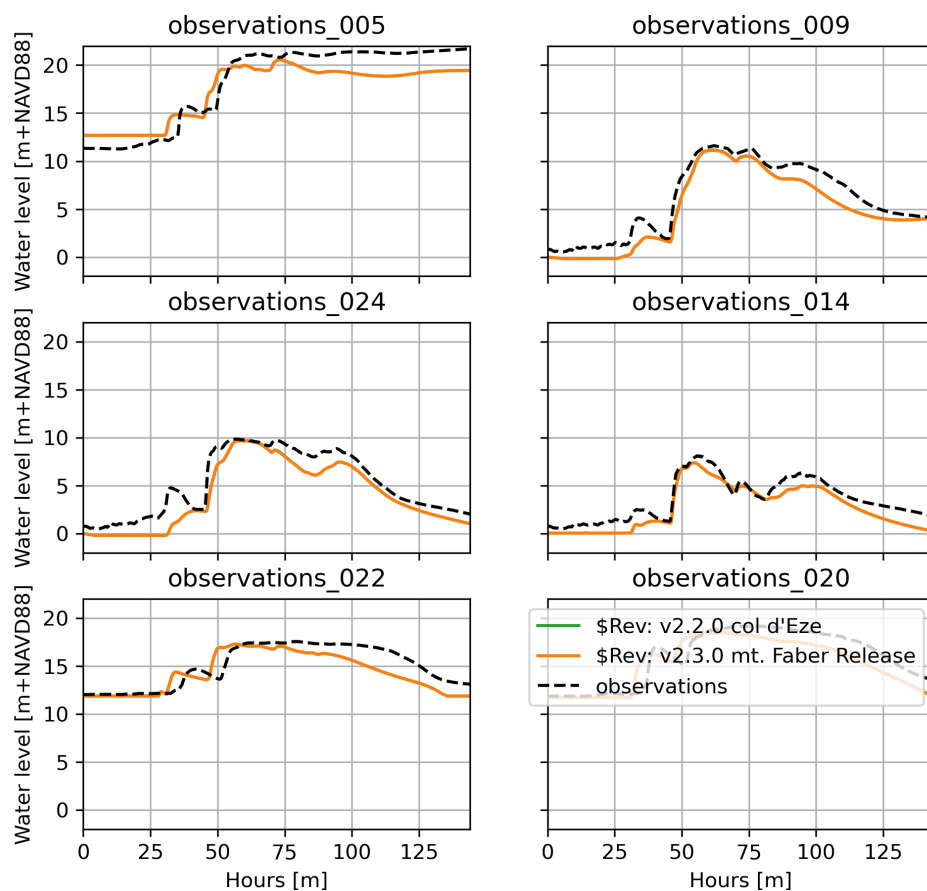


Figure 4.13: Time-series of observed and modeled (with stable and latest SFINCS release) water levels during Hurricane Harvey.

Table 4.8: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	1588.215	1101.594
RMSE loc:observations_005	-	0.001
RMSE loc:observations_009	-	0.012
RMSE loc:observations_024	-	0.013
RMSE loc:observations_014	-	0.004
RMSE loc:observations_022	-	0.001
RMSE loc:observations_020	-	0.001

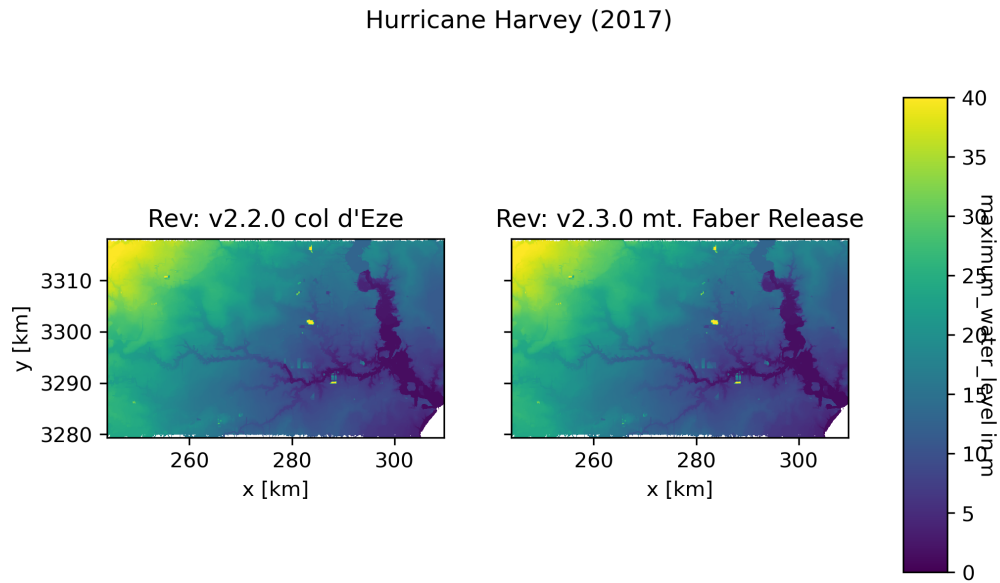


Figure 4.14: Maximum observed and modeled (with stable and latest SFINCS release) water levels during Hurricane Harvey.

4.4 Charleston during Hurricane Hugo (1989)

Experiment description

Test case of significant storm surge during Hurricane Hugo (1989) at Charleston, NC, USA.

Results

To assess the performance of the models, computed water levels are compared.

Charleston

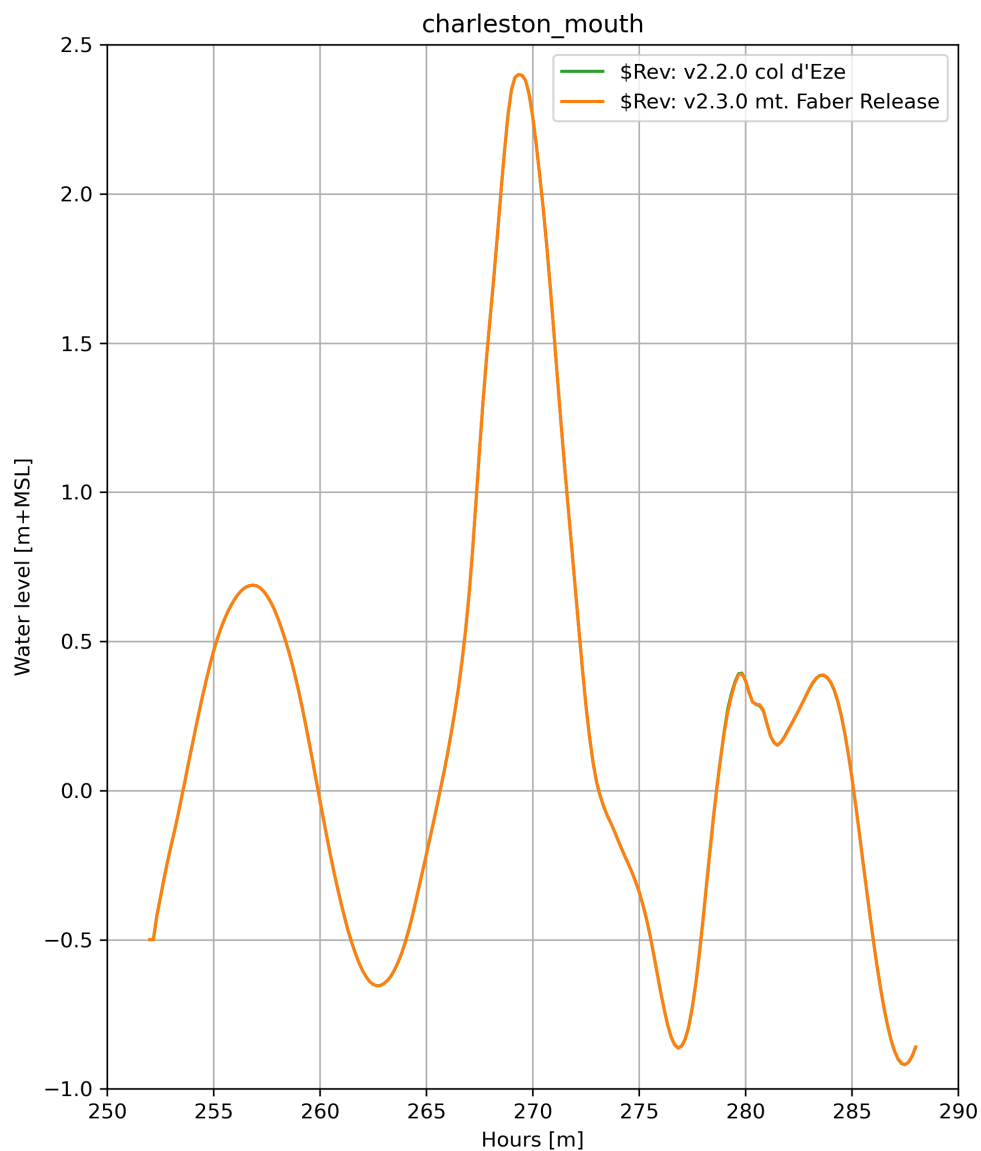


Figure 4.15: Time-series of modeled (with stable and latest SFINCS release) water levels during Hurricane Hugo.

Table 4.9: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:charleston_mouth
\$Rev: v2.2.0 col d'Eze	45.401	-
\$Rev: v2.3.0 mt. Faber Release	21.592	0.001

Charleston

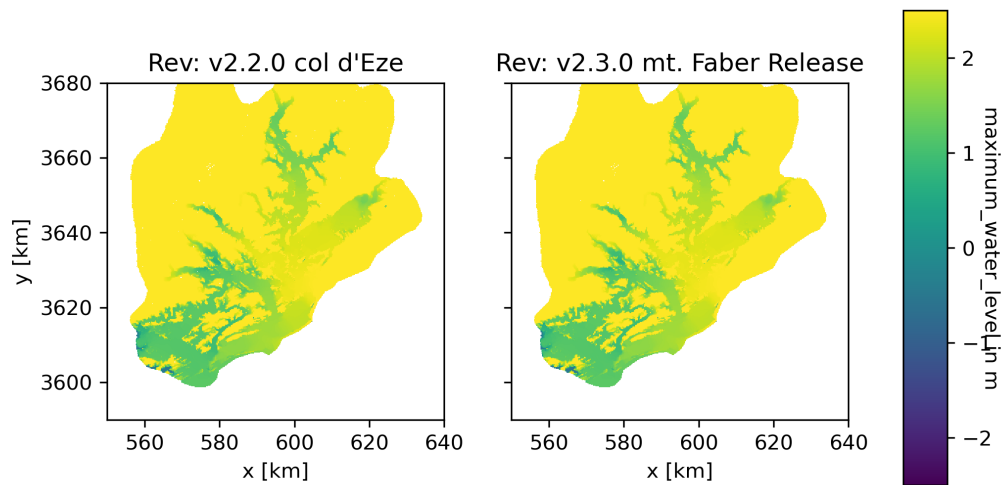


Figure 4.16: Maximum modeled (with stable and latest SFINCS release) water levels during Hurricane Hugo.

4.5 Bahamas during synthetic Hurricane

Experiment description

Test case of significant wind speeds during a synthetic hurricane at the Bahamas.

Results

To assess the performance of the models, computed wind speeds are compared.

Bahamas quadtree subgrid windspeed

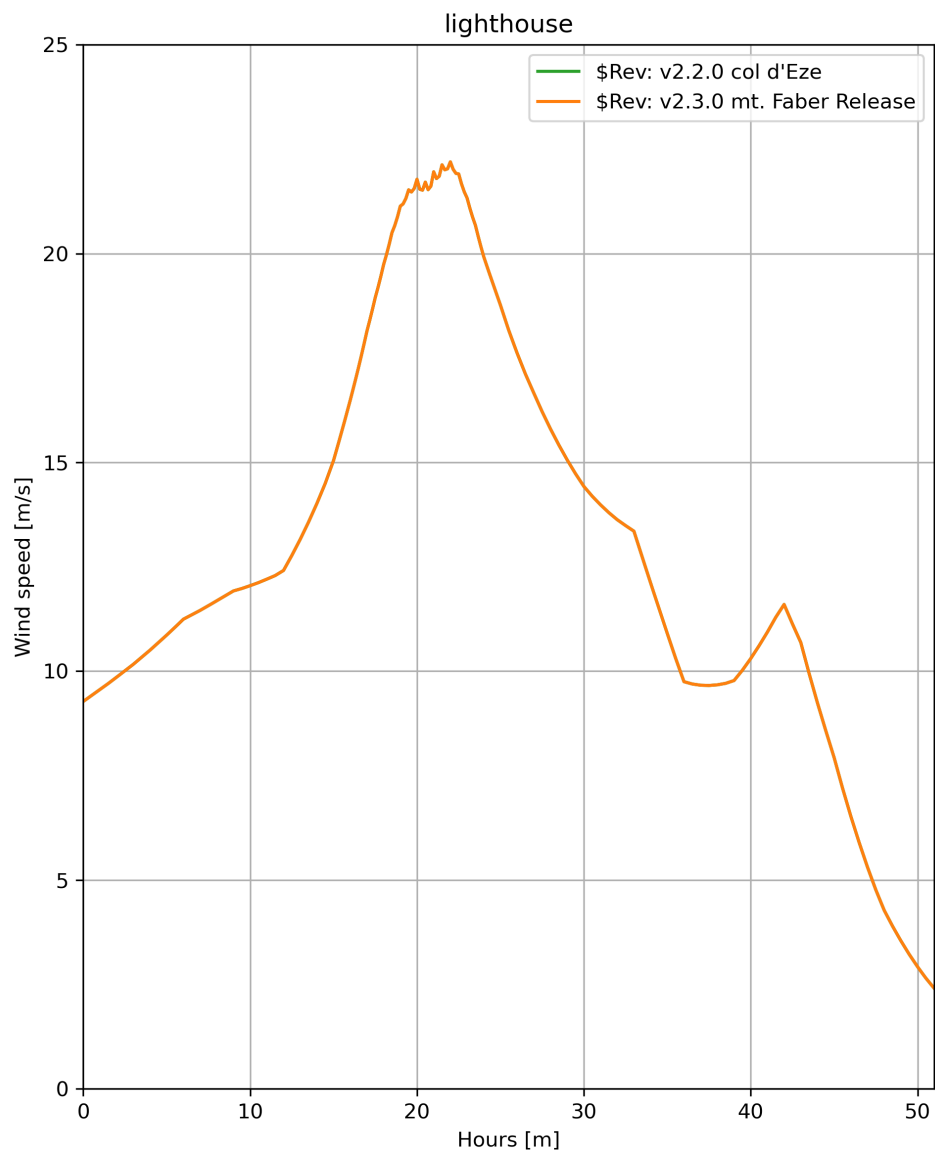


Figure 4.17: Time-series of modeled (with stable and latest SFINCS release) wind speeds.

Table 4.10: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:lighthouse
\$Rev: v2.2.0 col d'Eze	87.834	-
\$Rev: v2.3.0 mt. Faber Release	94.916	0.000

Results

To assess the performance of the models, computed water levels are compared.

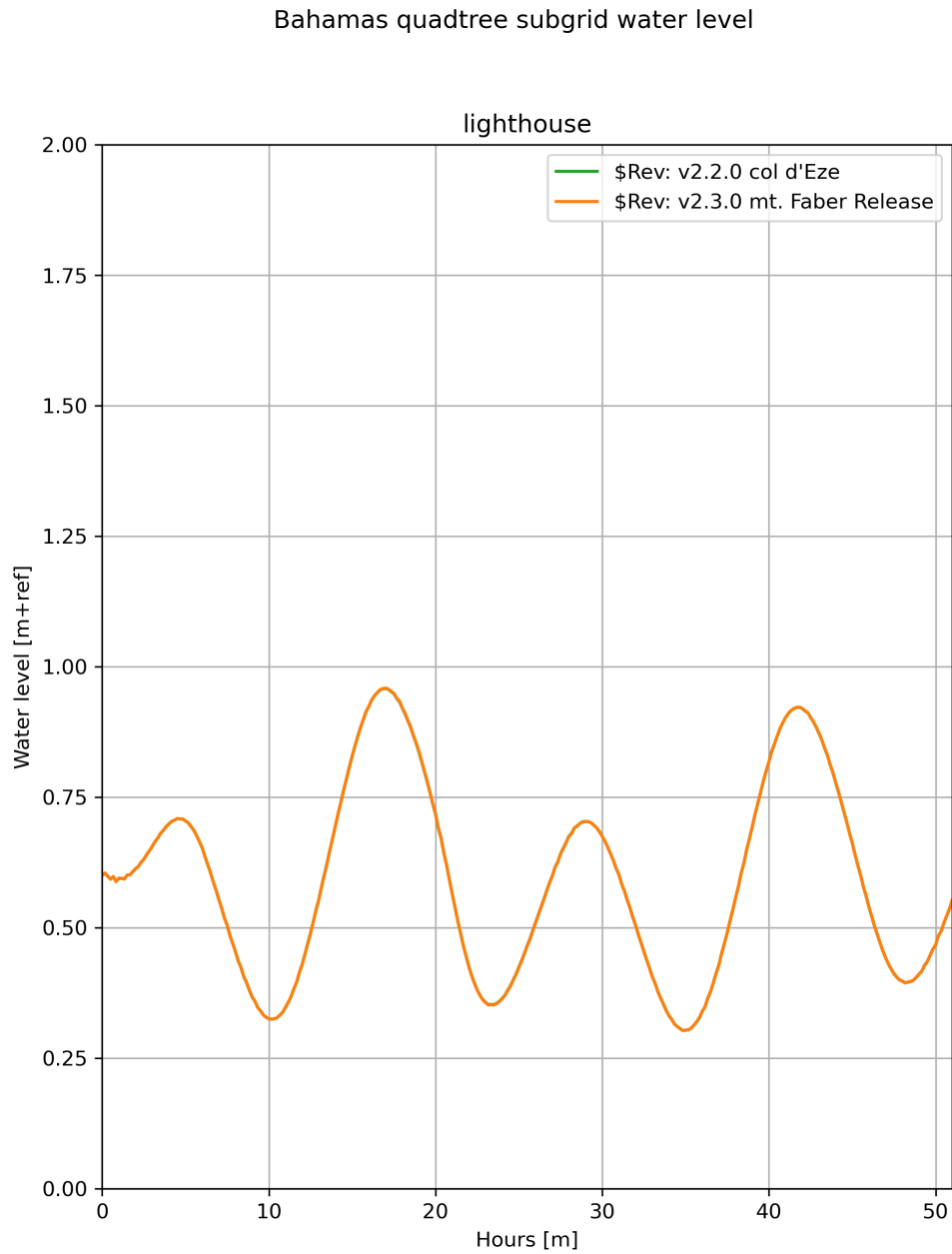


Figure 4.18: Time-series of modeled (with stable and latest SFINCS release) water levels.

Table 4.11: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:lighthouse
\$Rev: v2.2.0 col d'Eze	88.465	-
\$Rev: v2.3.0 mt. Faber Release	78.344	0.000

4.6 St. Johns river tide

Experiment description

Test case of pure tide propagation in the St. John's river, up to more than 100km upstream! The regular mode of FSINCS without subgrid is able to propagate the water levels far up-stream, but not completely upstream. The improvements with the subgrid mode is able to do this, visible in the tidal signal coming through at station 'WelakaStJohnsRiverFlorida' all the way upstream near Lake George.

Results - without subgrid

To assess the performance of the models, computed water levels are compared to observed time-series at observation stations of Xtide of locations along the St. Johns river using the regular SFINCS version.

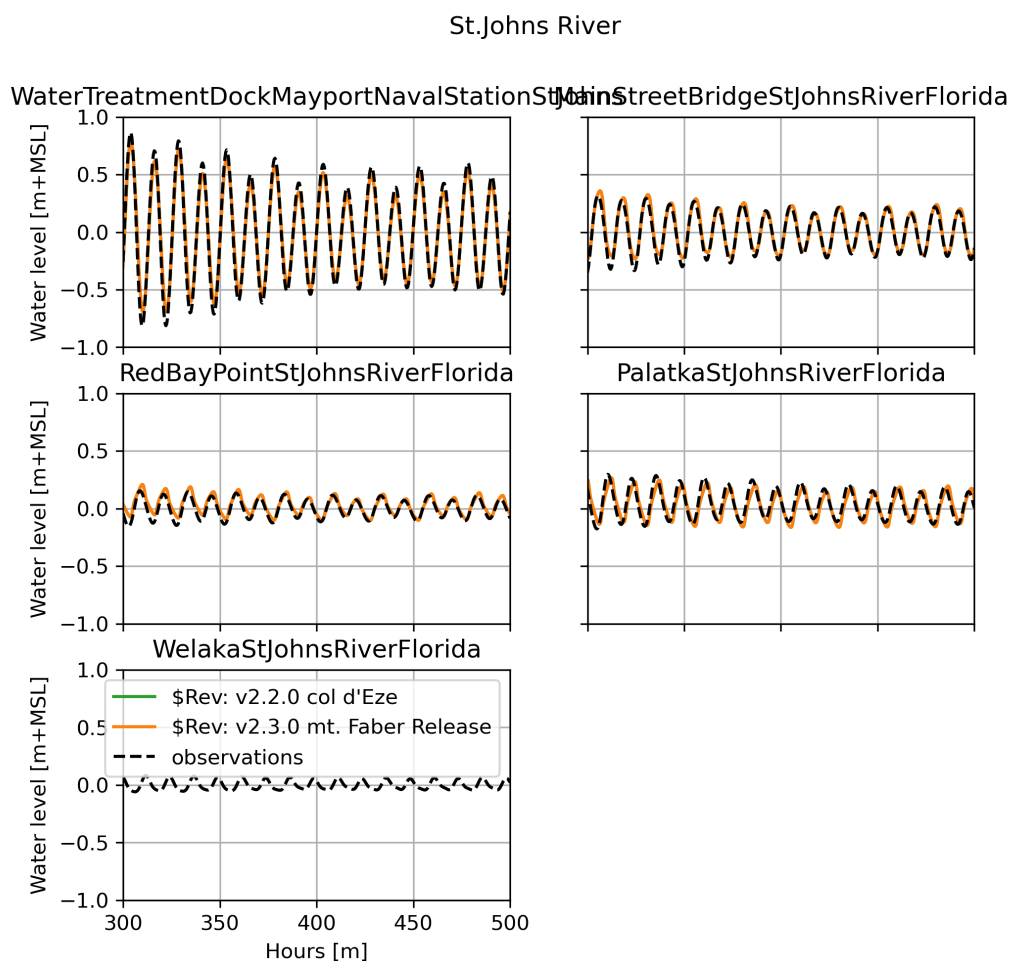


Figure 4.19: Time-series of observed and modeled (with stable and latest SFINCS release) water levels.

Table 4.12: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	582.762	580.632
RMSE loc:WaterTreatmentDockMayportNavalStationS...	-	0.000
RMSE loc:MainStreetBridgeStJohnsRiverFlorida	-	0.000
RMSE loc:RedBayPointStJohnsRiverFlorida	-	0.000
RMSE loc:PalatkaStJohnsRiverFlorida	-	0.000
RMSE loc:WelakaStJohnsRiverFlorida	-	0.000

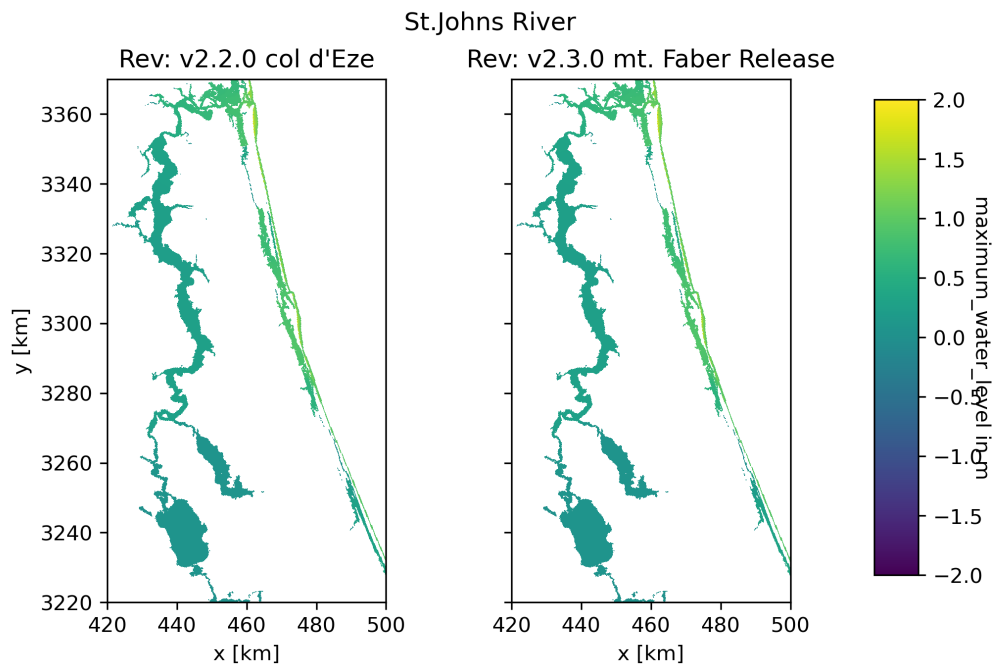


Figure 4.20: Maximum modeled (with stable and latest SFINCS release) water levels.

Results with subgrid

To assess the performance of the models with subgrid, computed water levels are compared to observed time-series at observation stations of Xtide of locations along the St. Johns river using the subgrid approach.

St.Johns River subgrid

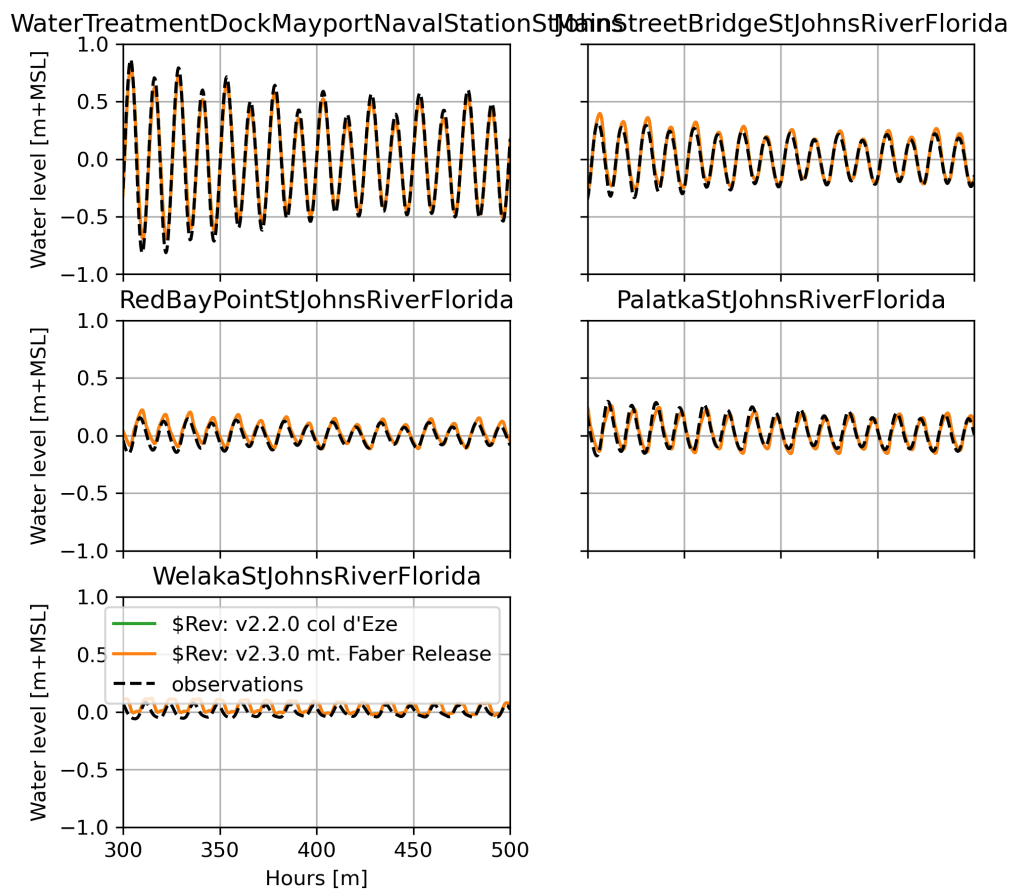


Figure 4.21: Time-series of observed and modeled (with stable and latest SFINCS release with subgrid) water levels during Hurricane Irma.

Table 4.13: Comparison in runtime and modelled water levels between latest release and stable release with subgrid.

Release version	\$Rev: v2.2.0 col d'Eze	\$Rev: v2.3.0 mt. Faber Release
Total runtime [s]	953.992	795.621
RMSE loc:WaterTreatmentDockMayportNavalStationS...	-	0.000
RMSE loc:MainStreetBridgeStJohnsRiverFlorida	-	0.000
RMSE loc:RedBayPointStJohnsRiverFlorida	-	0.000
RMSE loc:PalatkaStJohnsRiverFlorida	-	0.000
RMSE loc:WelakaStJohnsRiverFlorida	-	0.000

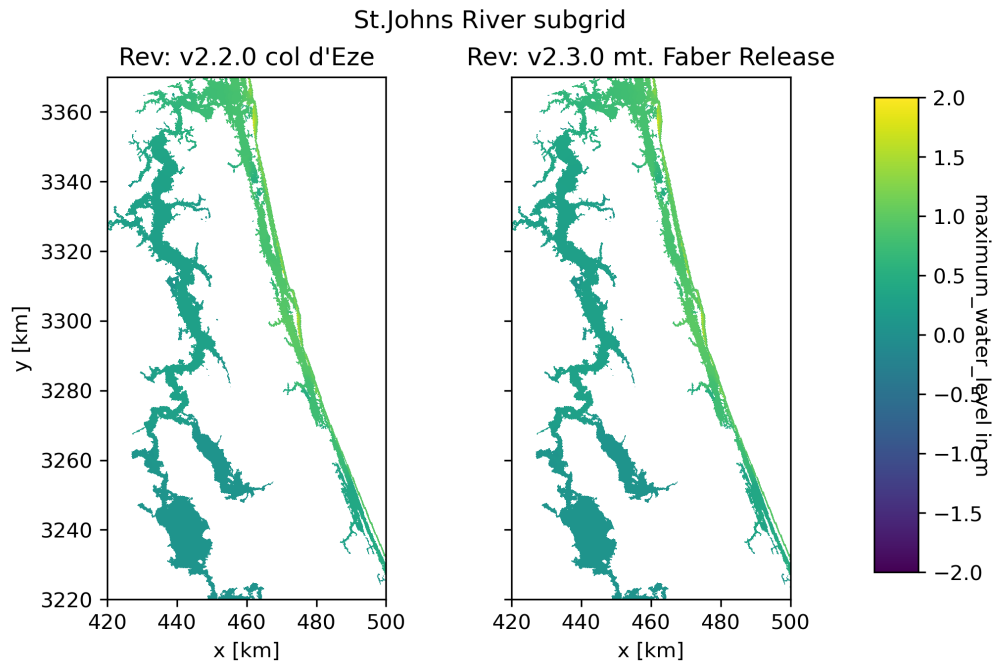


Figure 4.22: Maximum observed and modeled (with stable and latest SFINCS release with subgrid) water levels during Hurricane Irma.

4.7 Puget Sound

Results

To assess the performance of the models, computed water levels are compared for this model situated in the Puget Sound, Washington State, USA, during the flood event in the Nooksack river in 2009.

Ferndale, Puget Sound, 2009

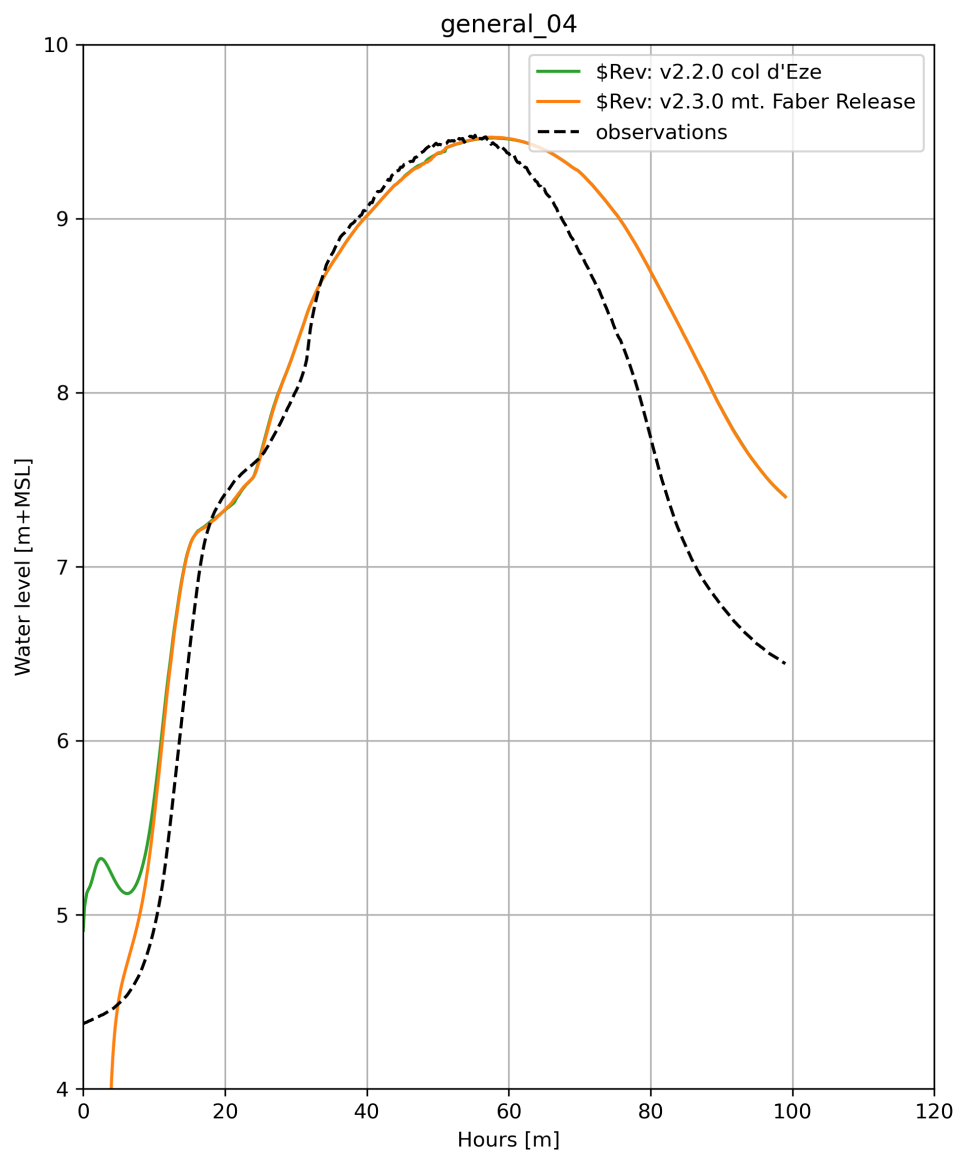


Figure 4.23: Time-series of modeled (with stable and latest SFINCS release) water levels.

Table 4.14: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:general_04
\$Rev: v2.2.0 col d'Eze	1317.749	-
\$Rev: v2.3.0 mt. Faber Release	1527.052	0.486

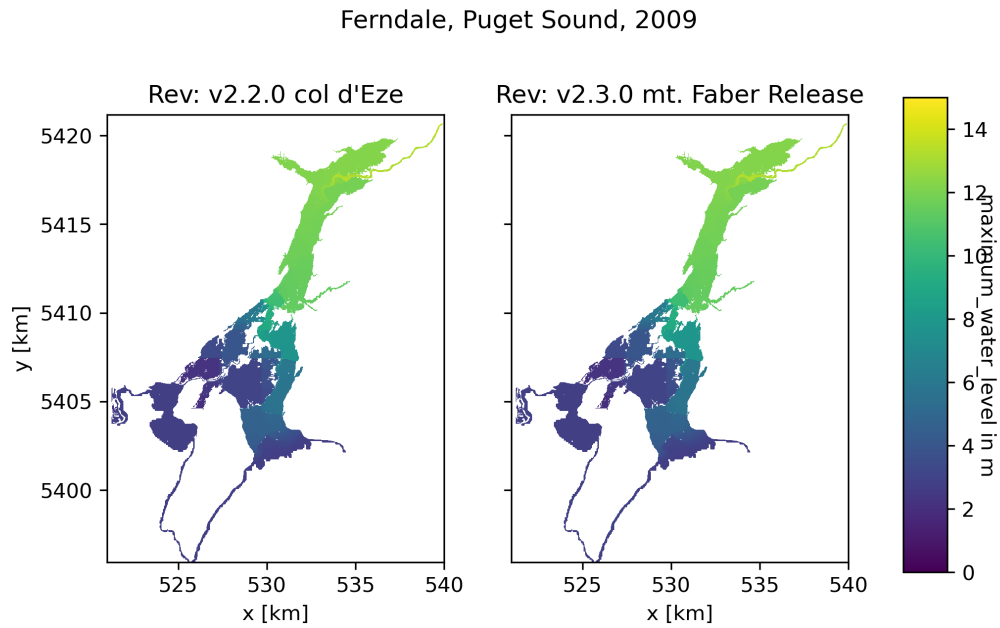


Figure 4.24: Maximum observed and modeled (with stable and latest SFINCS release) water levels.

Results

To assess the performance of the models, computed water levels are compared for this model situated in the Puget Sound, Washington State, USA, during the flood event in the Nooksack river in 2020.

Ferndale, Puget Sound, 2020

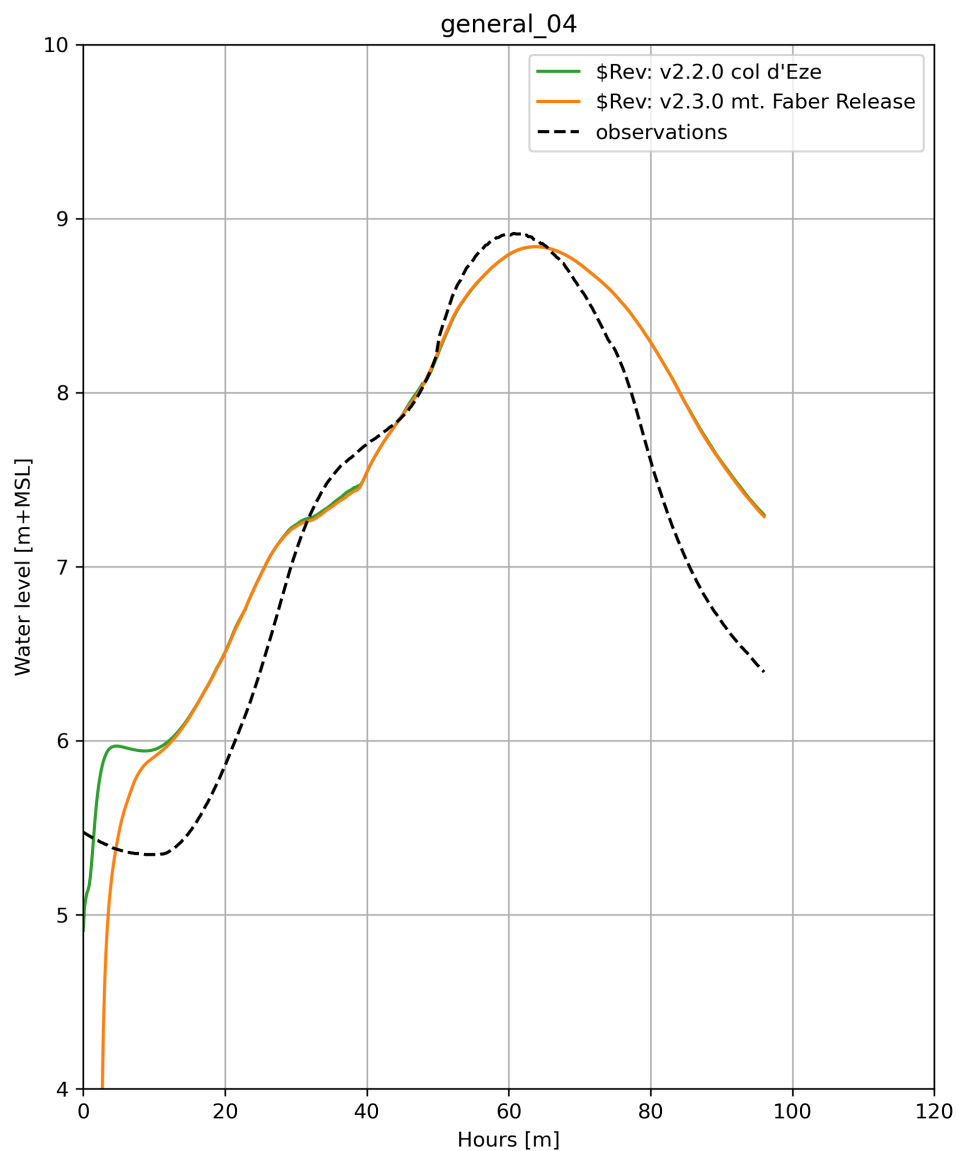


Figure 4.25: Time-series of modeled (with stable and latest SFINCS release) water levels.

Table 4.15: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:general_04
\$Rev: v2.2.0 col d'Eze	1089.708	-
\$Rev: v2.3.0 mt. Faber Release	1309.108	0.462

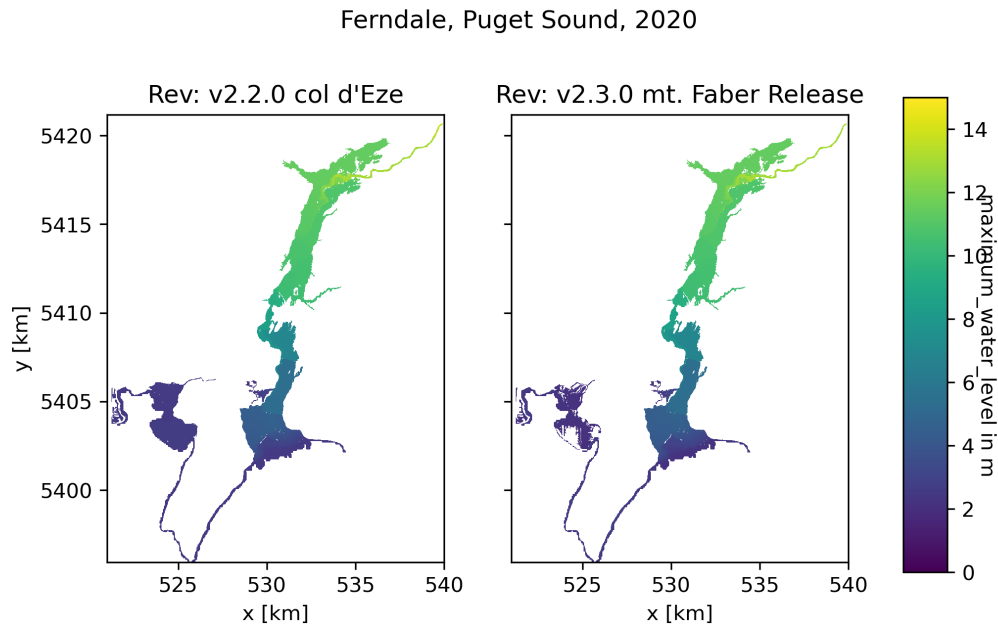


Figure 4.26: Maximum observed and modeled (with stable and latest SFINCS release) water levels.

4.8 Large scale TC model

Results

To assess the performance of the models, computed atmospheric pressure and maximum wind speeds are compared for this model situated in the SouthEast Coast of the US, being a large-scale spherical model. Compared is between the forcing using the netspwfile, compared to the same model forced with the same TC in the spwfile.

TC pressure in large-scale spherical model with netspwfile

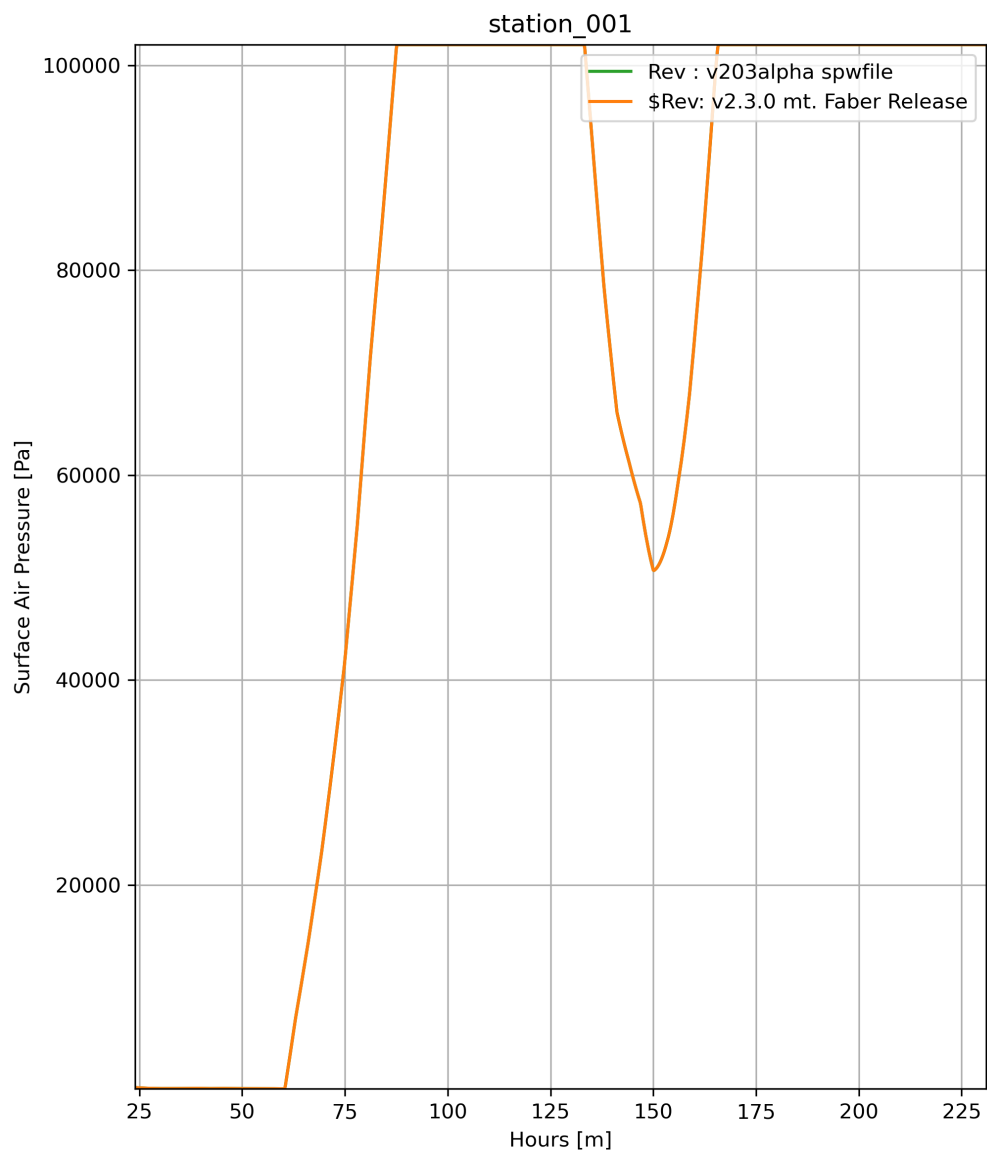


Figure 4.27: Time-series of modeled (with stable and latest SFINCS release) atmospheric pressure.

Table 4.16: Comparison in runtime and modelled atmospheric pressure between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:station_001
Rev : v203alpha spwfile	33.570	-
\$Rev: v2.3.0 mt. Faber Release	13.902	0.007

TC maximum windspeed in large-scale spherical model with netspwfile

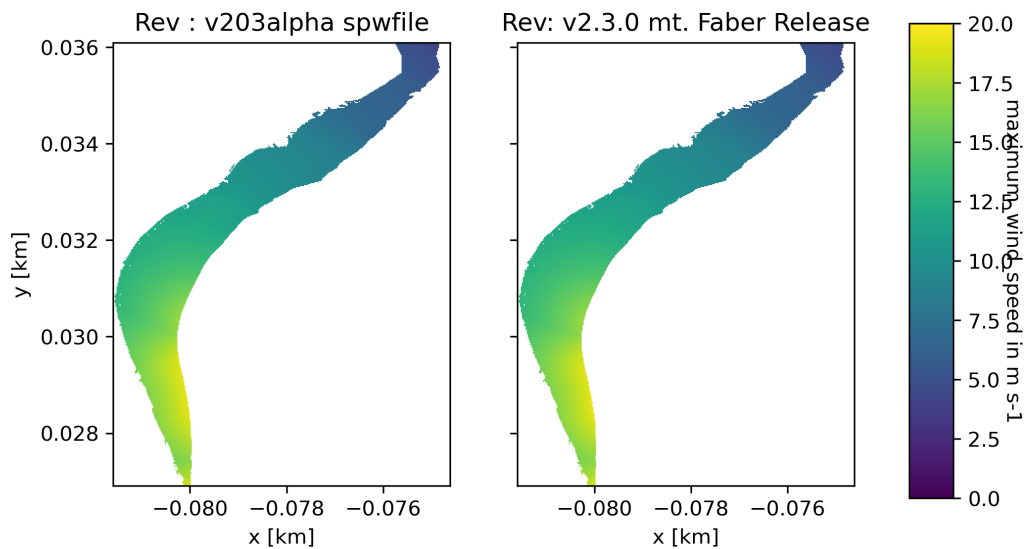


Figure 4.28: Maximum observed and modeled (with stable and latest SFINCS release) wind speeds.

4.9 Internal tide generation

Experiment description

Test case of the open HydroMT-SFINCS example notebook in Italy, with internal bcafile tidal generation within SFINCS.

Results

To assess the performance of the models, computed water levels are compared with the last SFINCS release.

internal tide generation test

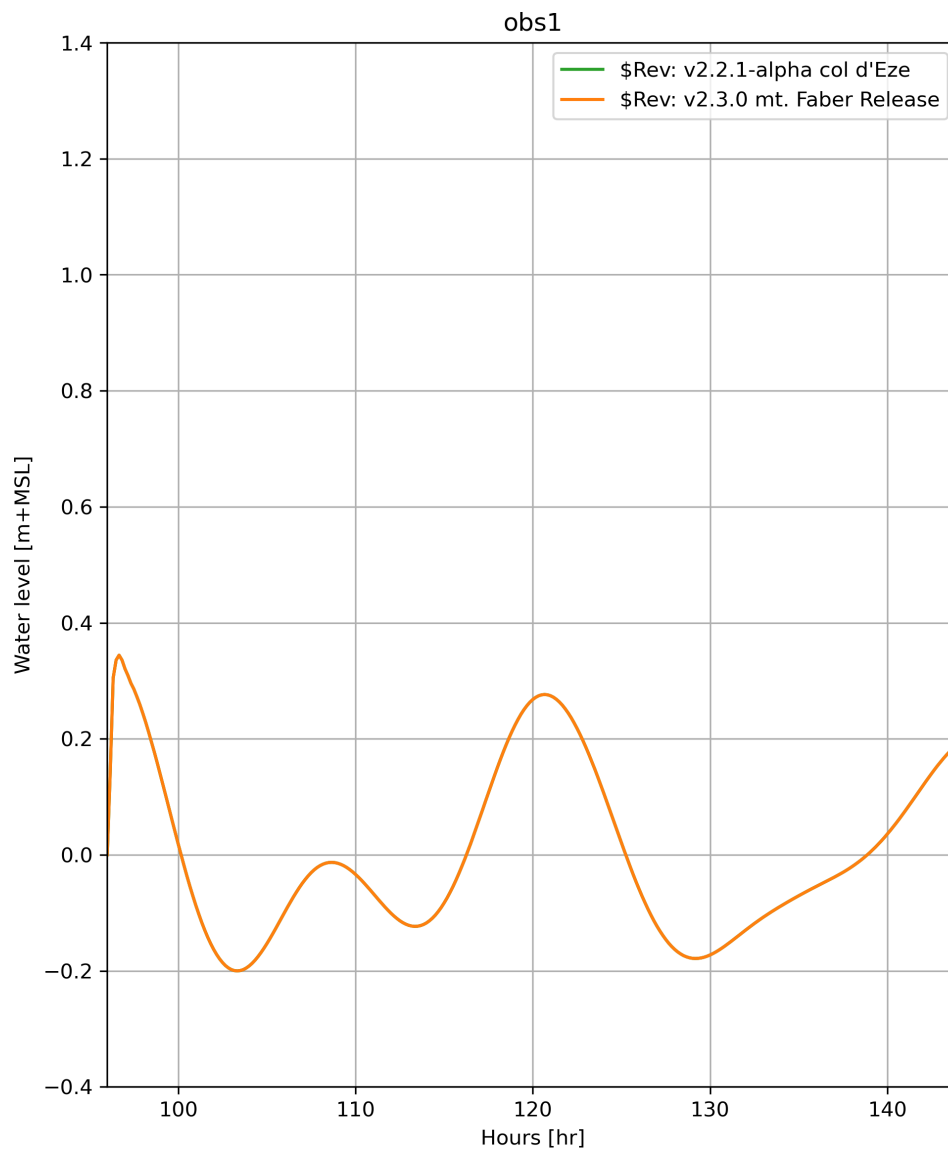


Figure 4.29: Time-series of modeled (with stable and latest SFINCS release) water levels.

Table 4.17: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:obs1
\$Rev: v2.2.1-alpha col d'Eze	5.032	-
\$Rev: v2.3.0 mt. Faber Release	0.570	0.000

Experiment description

Test case of the open HydroMT-SFINCS example notebook in Italy, with internal bcafile tidal generation within SFINCS superimposed with 1m surge from bzsfile.

Results

To assess the performance of the models, computed water levels are compared with the last SFINCS release.

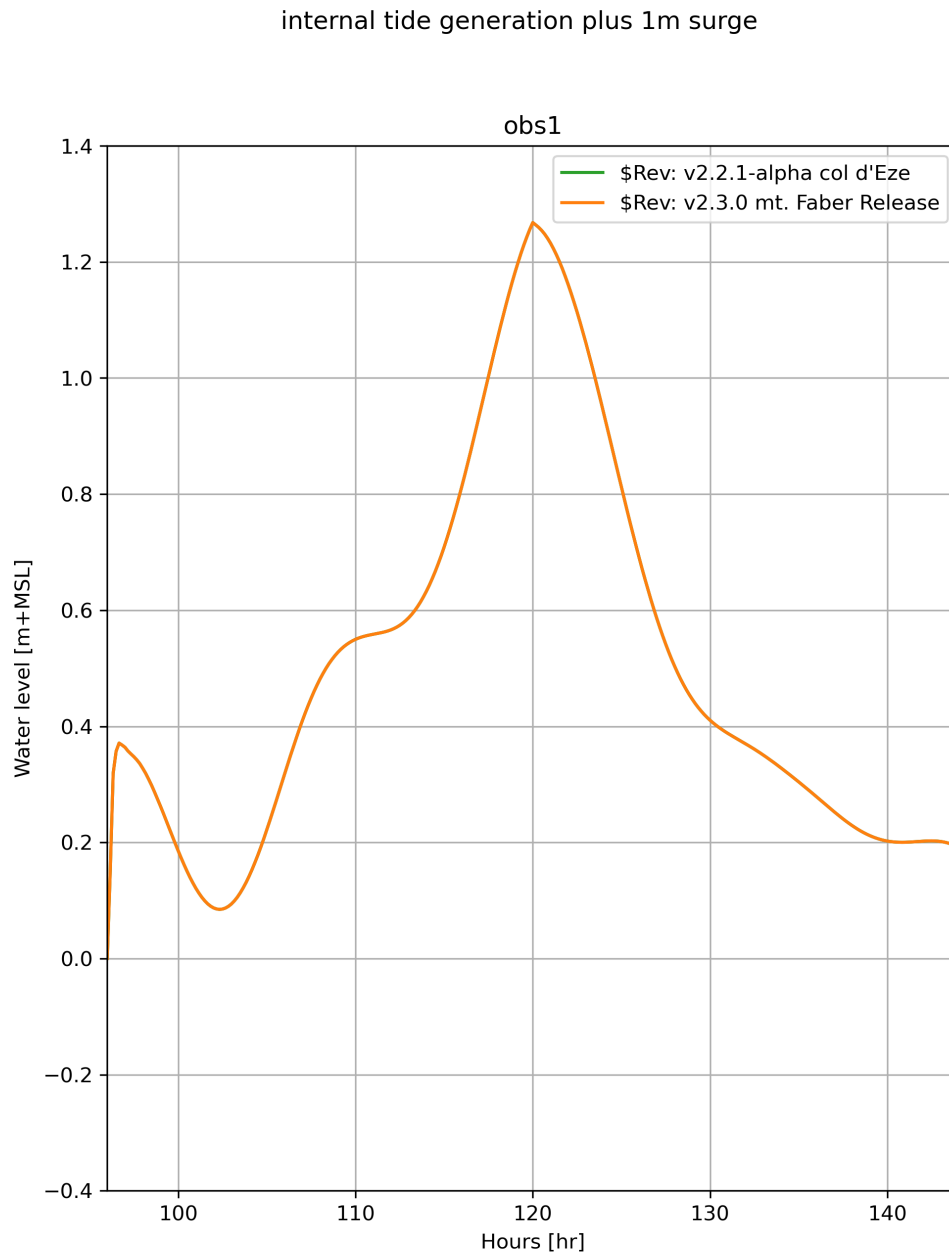


Figure 4.30: Time-series of modeled (with stable and latest SFINCS release) water levels.

Table 4.18: Comparison in runtime and modelled water levels between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:obs1
\$Rev: v2.2.1-alpha col d'Eze	5.177	-
\$Rev: v2.3.0 mt. Faber Release	0.722	0.000

4.10 Duck SnapWave IG model

Results

To assess the performance of the models, computed infragravity wave heights are compared for this model situated in Duck, NC, USA, during the DELILAH field experiment in 1990. As modelled in Leijnse et al. 2024.

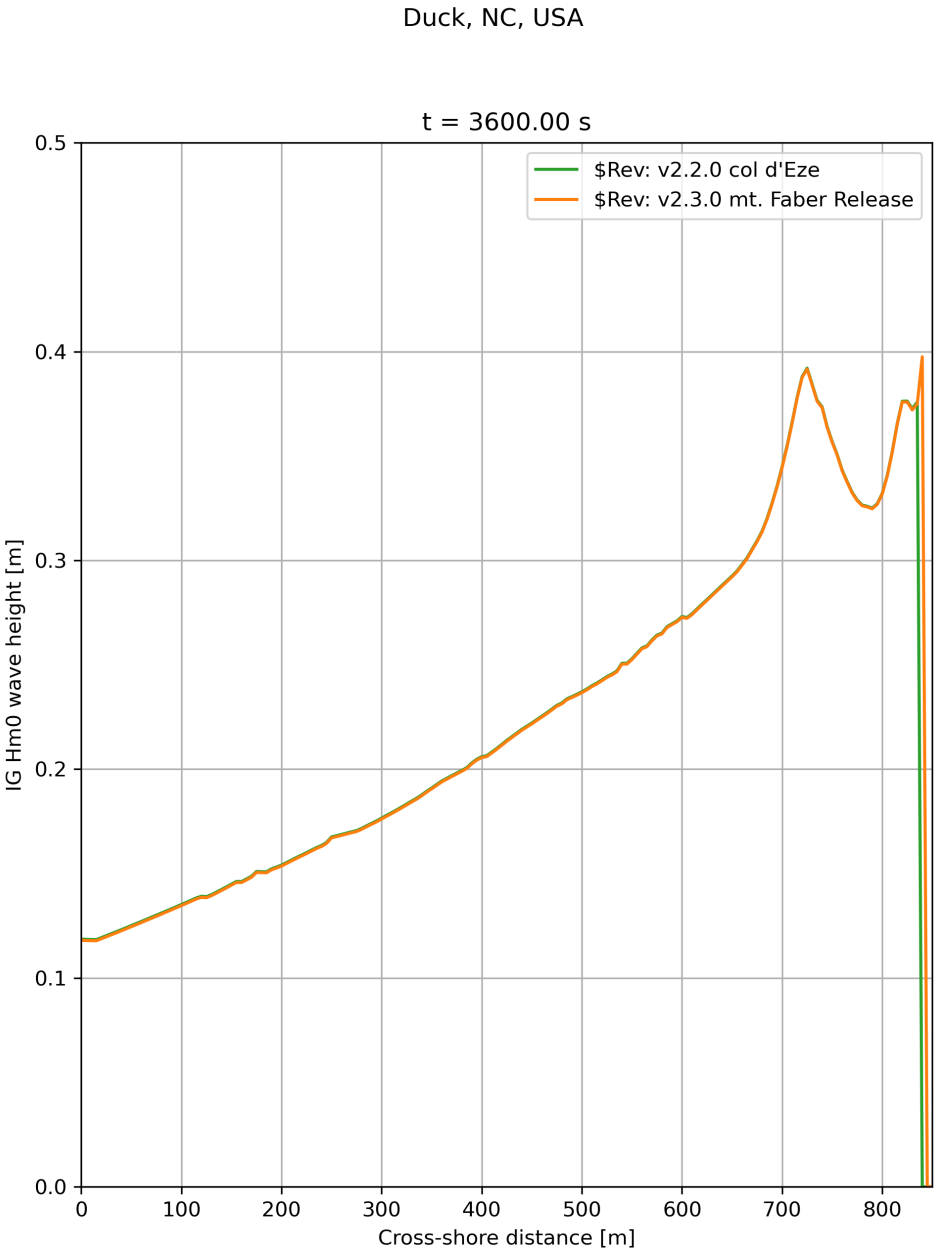


Figure 4.31: Cross-section of modeled (with stable and latest SFINCS release) infragravity wave heights.

Table 4.19: Comparison in runtime and modelled IG wave heights between latest release and stable release.

Release version	Total runtime [s]	RMSE t=3600.00s
\$Rev: v2.2.0 col d'Eze	40.125	-
\$Rev: v2.3.0 mt. Faber Release	31.567	0.030

Results

To assess the performance of the models, computed infragravity wave heights are compared for this model situated in Duck, NC, USA, during the DELILAH field experiment in 1990. As modelled in Leijnse et al. 2024, but then with a rising water level, that increased the H_{m0} wave height due to later breaking of waves.

Duck, NC, USA with rising water level

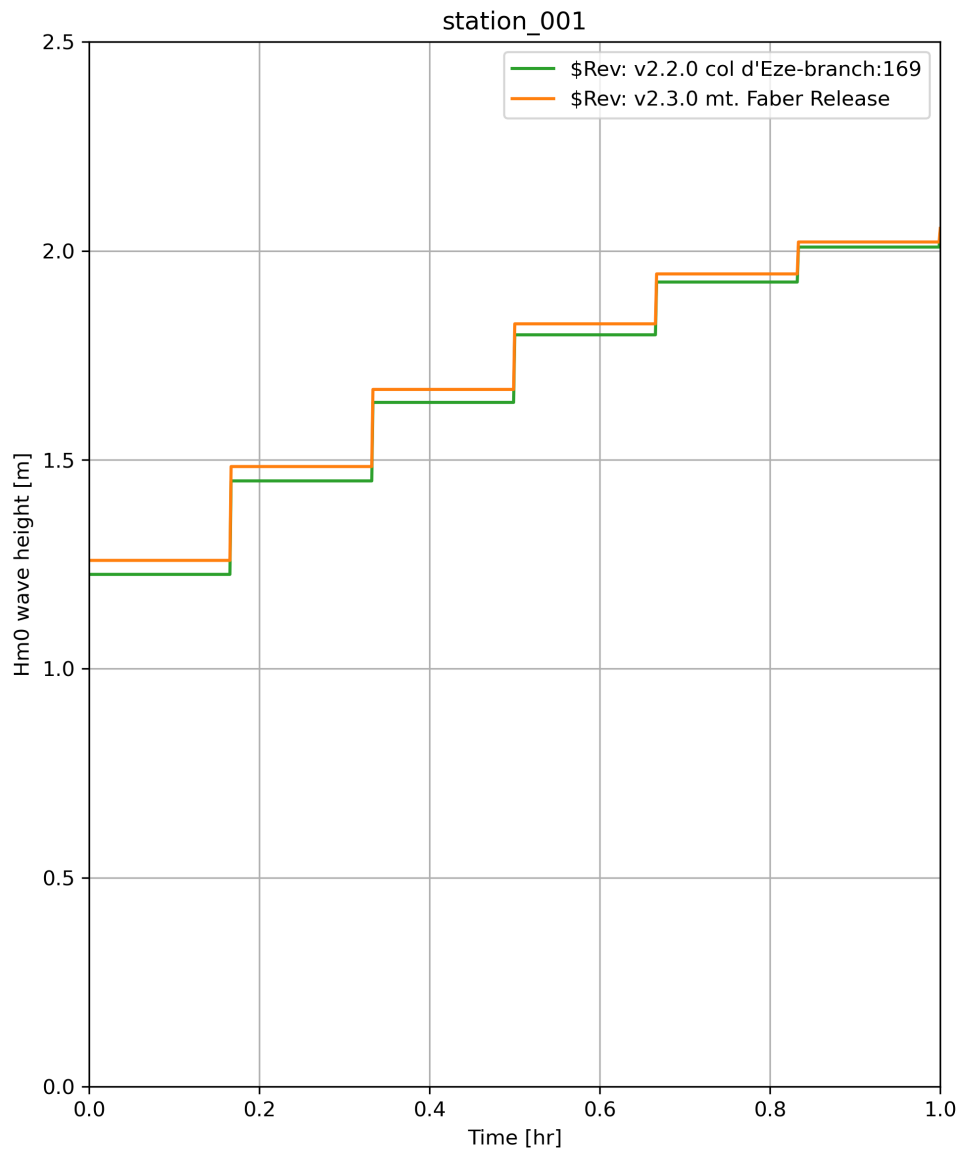


Figure 4.32: Cross-section of modeled (with stable and latest SFINCS release) infragravity wave heights.

Table 4.20: Comparison in runtime and modelled IG wave heights between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:station_001
\$Rev: v2.2.0 col d'Eze-branch:169	22.290	-
\$Rev: v2.3.0 mt. Faber Release	24.279	0.027

4.11 Outer Banks SFINCS quadtree SnapWave IG model

Results

To assess the performance of the models, computed infragravity wave heights are compared for this model situated in the Outerbanks at the North Carolina and Virginia, USA, coastlines for the wave conditions during the same 1990 wave conditions, based on ERA5 input data. As modelled in Leijnse et al. 2024.

Outer Banks, USA

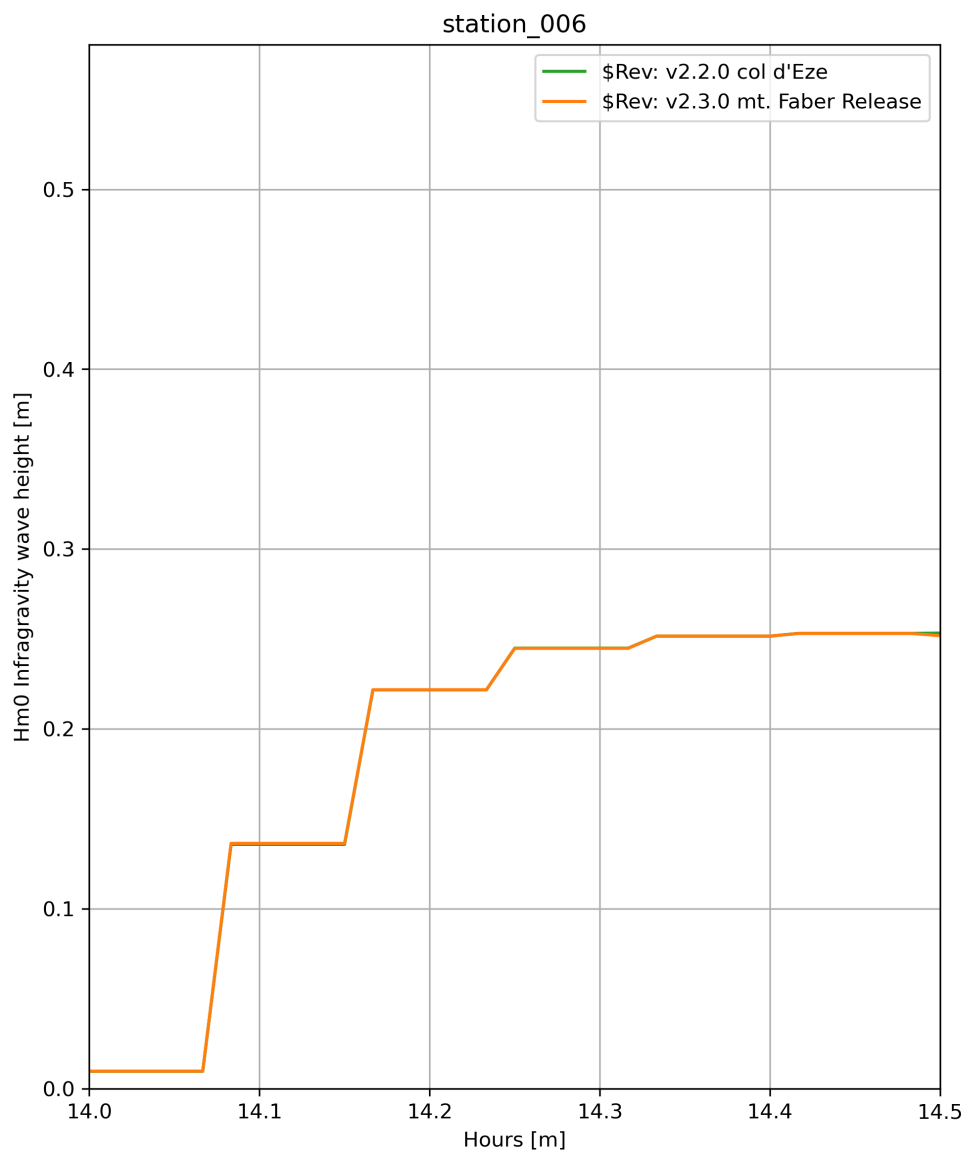


Figure 4.33: Overview of modeled (with stable and latest SFINCS release) infragravity wave heights.

Table 4.21: Comparison in runtime and modelled IG wave heights between latest release and stable release.

Release version	Total runtime [s]	RMSE loc:station_006
\$Rev: v2.2.0 col d'Eze	649.675	-
\$Rev: v2.3.0 mt. Faber Release	430.763	0.000

4.12 SFINCS release movie

Experiment description

Simulation of the SFINCS release movie with waves crashing over the embedded SFINCS logo.

Results

To assess the performance of the models, computed water depths are compared.

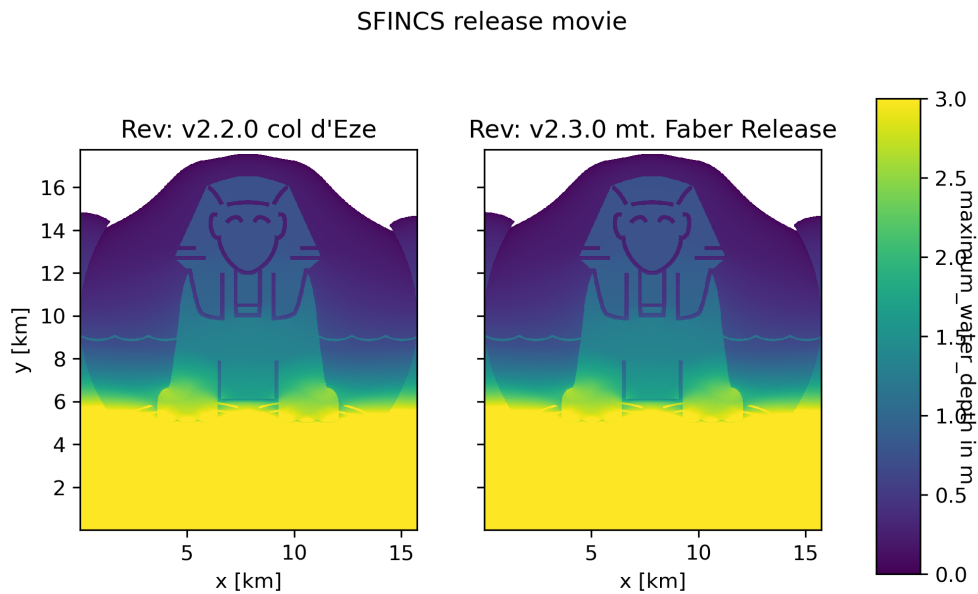


Figure 4.34: Maximum modeled (with stable and latest SFINCS release) water levels of the SFINCS logo.

4.13 SFINCS release movie 2023 version

Experiment description

Simulation of the SFINCS release movie with waves crashing over the embedded SFINCS logo, 2023 version.

Results

To assess the performance of the models, computed water depths are compared.

SFINCS release movie 2023

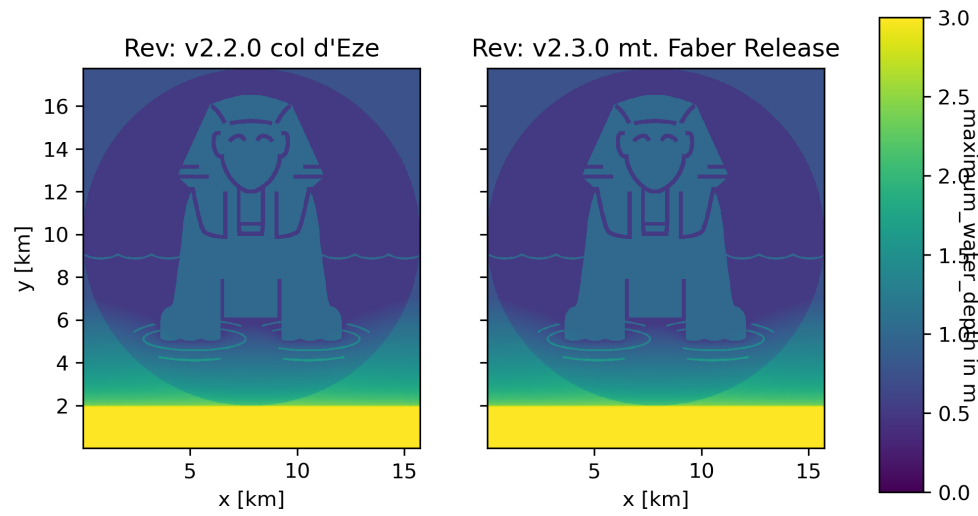


Figure 4.35: Maximum modeled (with stable and latest SFINCS release) water levels of the SFINCS logo.

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